Intermittent Core Formation in Herschel Gould Belt Filaments: a Sub-Classical *Local* Spacing with a Near-Classical Global Line Density*

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ABSTRACT

We re-analyse the spacing of dense cores along filament crests in four Herschel Gould Belt Survey clouds using Gaia DR3 distances. Four clouds are few, so the ranges quoted here are descriptive rather than formal confidence intervals, and characterise this nearby low-mass sample rather than a universal law. We first show that an all-pairs median separation does not estimate the local core spacing. For extent-filling distributions it converges to about 0.29 of the segment length, so it measures the size of the region selected. We therefore use the adjacent-core spacing along each skeleton and the direct core line density L/N. These disagree, and the disagreement is our main result. Cores occupy only 6–27 per cent of the skeleton length, counts in 1 pc cells are over-dispersed with Fano factors of 1.6–25, and the gap distribution shows a deficit of the smallest separations rather than clustering. Core formation is therefore intermittent, producing groups with short internal separations set within long stretches that have formed nothing. The local spacing is sub-classical by a factor of 2.4–4.3, whereas the global line density lies within about 1.8 of the classical Inutsuka & Miyama value. Athena++ calculations show that a contracting filament need not fragment at the equilibrium wavelength, but their isothermal growth-rate spectrum never turns over, so they bound the fragmentation scale at order the Jeans length rather than measuring it. Dynamical contraction, fibre projection and a force-balanced helical field remain untested against these data.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory demonstrated the ubiquity and importance of filamentary structure in nearby star-forming clouds (André et al. 2010), with a characteristic width of ∼0.1 pc across diverse environments (Arzoumanian et al. 2011), though the universality, and even the physical interpretation, of this scale is contested (Panopoulou et al. 2017, 2022), and a critical mass-per-unit-length $M_{\rm line,crit} \approx 16\ M_\odot\,{\rm pc}^{-1}$ (Ostriker 1964).

The observation. We quantify the core-to-core spacing homogeneously in four nearby HGBS clouds with Gaia DR3 distances (Zhang 2023). Throughout we keep three quantities distinct and do *not* equate them: the observed adjacent-core arclength spacing (hereafter adjacent-core spacing, ACS) statistic $s_{\rm ACS,med}$ (the median adjacent-core gap, a line-density scale, *not* itself a fragmentation wavelength); the fastest-growing fragmentation wavelength of our simulated filaments $\lambda_{\rm frag} \equiv \lambda_{\max}$ (since the growth-rate spectrum shows no resolved turnover, no maximum is actually resolved, so we quote the upper bound $\lambda_{\rm turn}$ throughout); and the classical Inutsuka–Miyama equilibrium value $\lambda_{\rm m} \approx 5$–$6\times$FWHM (Inutsuka & Miyama 1992; Fischera & Martin 2012). Two empirical results anchor the paper. First, the appropriate estimator is the ACS spacing: the widely-used all-pairs (pairwise) median is a region-extent-biased estimator *of local core spacing*, it validly estimates the segment extent but is inappropriate as a local-spacing estimator, converging to $\sim L/3$, the extent of whatever segment it is applied to (an individual filament, $L_{\rm fil} \sim 1$ pc, or a whole region), rather than to any local spacing (Section 2.6); pairwise medians, where they have been used in the HGBS literature, carry this bias. Second, on that estimator the observed conditional spacing $s_{\rm ACS,med}/W$ lies substantially below the equilibrium-cylinder wavelength $\lambda_{\rm m}$ under every skeleton and width convention. The deficit is a factor of roughly 2.4–4.3 below $\lambda_{\rm m}$ for the DisPerSE conventions, and further still under FilFinder; the residual magnitude is set by the skeleton algorithm and the width convention. The primary observed result is this convention *envelope*: $s_{\rm ACS,med}/W \simeq 0.8$–$1.9$ for the DisPerSE conventions, widening to $\simeq 0.4$–1.9 once the FilFinder skeleton is included (Fig. 9); one illustrative point within it is the region-Plummer/DisPerSE ACS-spacing ratio $(s_{\rm ACS,med}/W)_0 \approx 1.4$

* Both results are established for the nearby, low-mass Gould Belt sample ($N = 4$ clouds) and are not offered as a universal fragmentation law.

(Table 4), which we carry only as a bookkeeping reference convention, not the observed value. A reader's guide to all the λ/W values quoted in this paper is given in Table 2; the notational convention separating the observed statistic from the simulated wavelength is stated in Section 2.6.

Candidate explanations. Such a deficit relative to λ_m has been read as evidence for additional physics. Arzoumanian et al. (2019) established the $\sim$0.1 pc *width*, not a spacing; the sub-classical *spacing* and its usual fibre/magnetic reading come from the fragmentation literature (Fischera & Martin 2012; Zhang et al. 2020). Three mechanisms are candidates. (i) *Hierarchical fragmentation*, in which filaments are bundles of velocity-coherent fibres (Hacar et al. 2013, 2018; Yang et al. 2024), so that measuring core spacings across the bundle compresses the apparent filament-level spacing. (ii) *Magnetic tension* (Nakamura et al. 1993): only a *helical*/toroidal field shortens the axial mode. A static longitudinal field leaves it unchanged and a perpendicular field lengthens it (Section 3.3), so a magnetic route to a shorter spacing is a specifically helical effect. Its force-balanced case is untested here, and we deliberately leave the helical/toroidal magnetic mechanism as an identified open problem that our MHD campaign does not close (Section 4.3). (iii) *Dynamical fragmentation*: a still-collapsing filament fragments at a scale set by its current, contracting state rather than by the marginally-stable equilibrium mode.

That real, still-accreting filaments need not fragment at the equilibrium wavelength is already qualitatively expected in the dynamical-ISM paradigm, in which filaments form and evolve out of equilibrium while accreting (André et al. 2014; Clarke et al. 2016, 2017, 2020). Our aim is therefore not to propose a new mechanism but to quantify the observation and test whether the simplest dynamical reading is even viable. The genuinely new contributions are three: (i) a homogeneous, Gaia-DR3, ACS-corrected, multi-convention observational quantification of the spacing across the four primary clouds; (ii) identifying the region-extent ($\approx 0.293\,L$) bias as the source of the discrepancy in the HGBS pairwise-median literature; the underlying triangular-distribution order statistic is elementary and textbook, and region-extent biases of this kind are already familiar in spatial point-process statistics (the same Clark–Evans nearest-neighbour context used in Section 2.10), so we claim no new mathematics, only the identification of this estimator bias in the filament-spacing context; and (iii) a specific longitudinal near-critical MHD simulation that fragments sub-classically, with an equilibrium-recovery control that recovers the classical wavelength under the identical pipeline.

We present the full-coverage ACS analysis for the four primary HGBS regions (Orion B, Aquila, Perseus, Taurus; the eight-region sample is retained only for the legacy pairwise-median statistic), followed by the simulation campaign, its validation and the interpretation. Two groups of statements should be kept apart. *Established here:* the pairwise median is a region-extent-biased estimator of local core spacing (converging to $L/3$); the ACS spacing is the appropriate observable; the spacing is sub-classical in *sign* under every convention, while its magnitude is convention-dominated, the skeleton choice being the single largest term. *Not established:* that dynamical collapse is the dominant cause, that hierarchical fibres are unimportant, or that magnetic tension is excluded. Symbols are collected in Table 1. With only four independent clouds, these results characterise the nearby Gould-Belt sample alone, not a universal fragmentation law.

Table 1. Principal symbols.

Symbol	Meaning
$s_\mathrm{ACS,med}$	observed adjacent-core spacing (ACS) statistic (median adjacent gap; a line-density scale, *not* a resonant/fragmentation wavelength; cf. Section 2.10)
W_fil	observed filament FWHM (beam-deconvolved Plummer; the reference width)
W_core	simulation initial Gaussian half-width σ ($0.3\,\lambda_\mathrm{J}$)
W_form	FWHM of the evolved, projected simulation profile
$T_1 \equiv \eta_W$	$W_\mathrm{form}/W_\mathrm{fil}$, the simulation-to-observation width normalisation (forward-model band 0.59–0.78; adopted 0.61)
λ_J	background Jeans length (= 1 in code units)
$\lambda_\mathrm{max} \equiv \lambda_\mathrm{frag}$	simulated fastest-growing fragmentation wavelength; *upper bound; no resolved peak for contracting runs* ($\lesssim 1.1\,\lambda_\mathrm{J}$; the equilibrium-control peak *is* resolved, at $22H$), so we quote λ_turn in its place
λ_turn	upper bound on the simulated fastest-growing wavelength (the turnover is not resolved; $\lesssim 1.1\,\lambda_\mathrm{J}$); the quoted stand-in for λ_max
$\lambda_\mathrm{m} \equiv \lambda_\mathrm{IM92}$	classical Inutsuka–Miyama equilibrium-cylinder wavelength $\approx$ 5–6 W_FWHM (= 4× flat diameter)
$\Gamma(\lambda)$	linear growth rate of axial mode of wavelength λ
f	thermal line-mass fraction $M_\mathrm{line}/M_\mathrm{line,crit}$
β	plasma beta $8\pi P/B^2$
$\mathcal{M}$	amplitude label of the seed perturbation ($\delta v = \mathcal{M} c_s \times 10^{-4}$; a linear seed, *not* a turbulent Mach number; driven transonic turbulence is tested separately in Appendix F3)
$\mu_\Phi/\mu_{\Phi,\mathrm{crit}}$	normalised mass-to-flux ratio
θ	angle between field and filament axis
N_J	cells per local Jeans length (Truelove criterion)
C	density contrast $\rho_\mathrm{max}/\langle\rho\rangle$

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample comprises 8 high-confidence HGBS regions (four *primary*, Table 3; four *limited*, Table A1 in Appendix A). The original catalogues (Könyves et al. 2015; Könyves et al. 2020; Ladjelate et al. 2020; Pezzuto et al. 2021) adopted distances of 130–400 pc from the best estimates available in 2010–2015 (Table 11 lists the pre-Gaia distance and its source per region; note that Könyves et al. 2020 already adopted ~ 400 pc for Orion B); Gaia DR3 parallaxes (Zhang 2023) have since enabled substantial revisions, largest for the more distant regions, whose pre-Gaia distances were the most uncertain.

We adopt the Gaia DR3 YSO-based distances from Zhang (2023) (Table 4), with typical uncertainties of 3%–5%. The revisions are strongly region-dependent (largest for Serpens 260 $\rightarrow$ 458 pc, +76%, and Aquila 260 $\rightarrow$ 436 pc, +68%; Orion B, Taurus and Ophiuchus essentially unchanged); since spacings scale linearly with distance, the core-count-weighted

Table 2. Reader's guide to the λ/W values quoted in this paper. All observational entries are $s_{\mathrm{ACS,med}}/W$ or $(L/N)/W$ line-density statistics, not fragmentation wavelengths; see Section 2.8. Ranges are descriptive, not confidence intervals.

Quantity	λ/W
All-pairs pairwise median (biased; not a spacing, $\approx 0.293L$)	≈ 2.8
Median adjacent ACS, region-Plummer (adopted reference)	≈ 1.4
Median adjacent ACS, fixed 0.10 pc	≈ 1.9–2.1
Median adjacent ACS, FilFinder	≈ 0.4–0.8
Per-segment local-width $\Delta s/W_{\mathrm{local}}$ (median)	≈ 0.6–1.1
Direct line density $(L/N)/W$ (fixed 0.10 pc)	≈ 3.4–5.8
Mean adjacent, fixed 0.10 pc (branch-jump inflated)	≈ 2.1–4.7
Simulation upper bound (matched Plummer)	≈ 1.0–1.3
Classical Inutsuka–Miyama	≈ 5–6

Table 3. Primary HGBS sample with Gaia DR3 distances. Per-region spacings are pairwise-median values rounded to three significant figures; the core-count-weighted mean of the four primary regions is 0.290 pc. The four *limited* regions (Ophiuchus, Serpens, TMC1, CRA) and the full 8-region weighted mean (0.279 pc) are listed in Appendix A, Table A1.

Region	D (pc)	N_{core}	Pairwise med. (pc)	S.E. (pc)	Status
Orion B	386	1,844	0.313	0.047	Primary
Aquila	436	749	0.346	0.047	Primary
Perseus	300	816	0.248	0.040	Primary
Taurus	135	536	0.198	0.040	Primary
Weighted Mean[†]		**3,945**	**0.290**	,	

[†]Core-count-weighted mean of the four primary regions. A formal core-count-weighted standard error (treating cores as independent) is *not* the relevant uncertainty, since the four clouds are the independent units (Section 2.11); the cloud-to-cloud scatter is carried instead.

mean rises only modestly ($\sim 10\%$ for the primary regions), dominated by Aquila rather than being a uniform +32% shift. Because Gaia DR3 thus compresses or expands each cloud non-uniformly, the revision is not a single global rescaling, which is a further reason the clouds must be treated as $N = 4$ independent units rather than by pooling cores that carry very different distance corrections. The observed spacing is a line-density statistic, not a resonance, and coincides with the theoretical wavelength only to order of magnitude.

The analysis includes all HGBS-identified cores (starless, prestellar and protostellar), extracted and classified within the consistent HGBS framework (Könyves et al. 2015; Könyves et al. 2020). Including unbound starless and protostellar sources may introduce some bias, but fragmentation produces a continuum of boundness states, and restricting the sample to prestellar cores would shrink several regions (notably TMC1, CRA and Serpens) below statistical usefulness. Core positions and properties are taken directly from the published catalogues.

Protostellar migration displacements of 0.01–0.05 pc (Kirk et al. 2016; Mattern et al. 2018) are small against the ~ 0.2–0.3 pc spacing; we estimate the resulting ACS bias at ~ 5–10% and carry it as a systematic (Section 2.11).

2.2 Measurement Methodology

The analysis pipeline, from raw HGBS maps to the final characteristic $s_{\mathrm{ACS,med}}/W$, is summarised in Fig. 1. Filament skeletons were identified using DisPerSE (Sousbie 2011) following standard HGBS methodology (Arzoumanian et al. 2011), with a 3σ persistence threshold and manual editing restricted to removing obvious artefacts (e.g. spurs connecting unrelated structures). Its effect on the spacing is small ($\lesssim 5\%$), bounded by the junction-retention and association-threshold sensitivity tests as a *proxy* (Table 8); a dedicated $2\sigma/4\sigma$ persistence-threshold scan and a blind, multi-editor re-skeletonisation are left as future work (Section 6.4). Because skeletons and core positions derive from distance-independent angular positions, the Gaia DR3 revisions require no re-extraction. The measured spacing is insensitive to these choices: excluding junctions changes it by $< 5\%$ and varying the association threshold between $1\times$ and $1.5\times$ the width by $< 3\%$ (a coarse two-point check, distinct from the finer 0.04–0.08 pc half-width sweep of Section 2.7 that gives a 4–7% per-region spread; the primary ACS pipeline associates cores within a half-width, ~ 0.06 pc, of a spine).

Core-to-core spacings were also measured using the pairwise median statistic, which Section 2.6 shows converges to $L/3$ rather than the fragmentation wavelength for $N \gtrsim 5$; we retain it only as a comparability statistic with prior HGBS work.

Because physical spacing is angular separation times distance, the $\lambda \propto d$ scaling is tautological and we do not present it as corroboration; the relevant check is that the *residuals* about that scaling show no trend with the absolute revision fraction $|\Delta D/D_{\mathrm{original}}|$, the two most-revised clouds (Aquila, Serpens) do not stand out as having the largest or smallest residuals, so the revision does not preferentially bias the most-revised clouds. With only four clouds a formal correlation test has negligible power and we do not quote one. Two caveats remain, and we flag them as limitations rather than resolve them: the YSO-based distances (Zhang 2023) are assumed to apply to the dense gas measured here, and the Aquila/Serpens Rift is known to have substantial line-of-sight depth, so assigning a single distance may blend populations and inflate the inferred spacing. As an order-of-magnitude estimate for Aquila, a Rift depth of order tens of pc (the Serpens/Aquila complex spans $\sim$ tens of pc in line of sight in the 3-D dust reconstructions of Zucker et al. 2020 and Green et al. 2024) set against a projected spacing ~ 0.2 pc allows chance line-of-sight superposition. We have quantified this with a geometric Monte-Carlo model in which two crests at different line-of-sight depths are projected onto a common spine, with intrinsic spacing 0.20 pc at $d = 436$ pc. Superposition interleaves the two chains and therefore *shortens* the measured median: for depths of 10–60 pc the recovered median falls to 0.41–0.49 of the true value. The effect of the distance spread alone, which rescales each spacing by $d/(d+z)$, is far smaller, ± 1 to ± 7 per cent over the same depth range. Blending thus biases the ACS towards *more* sub-classical values, not less, and the ~ 50 per cent figure is a worst case in which two full crests overlap; the realistic effect is much smaller because

(i) HGBS column-density maps (250 μm reference) + Gaia DR3 distances

(ii) DisPerSE skeleton extraction (3σ persistence threshold)

(iii) Morphological closing (8 iterations) + re-skeletonisation (bridge fragmented segments; components ≥ 40 px)

(iv) Core association: cores within 0.06 pc of a spine are assigned; others rejected

(v) Graph ordering: cores ordered by double-BFS arclength along each component

(vi) Adjacent-core ACS spacing (in pc, via Gaia DR3 distance)

(vii) Injection–recovery bias correction ($\div$ 1.11–1.29, region-dependent)

(viii) Characteristic $s_{\rm ACS,med}/W$ = bias-corrected median spacing / $W_{\rm fil}$

Figure 1. Analysis pipeline from raw HGBS maps to the final characteristic $s_{\rm ACS,med}/W$ (a line-density statistic, not a wavelength). Each step's parameters are given in the text (Sections 2.6–2.7).

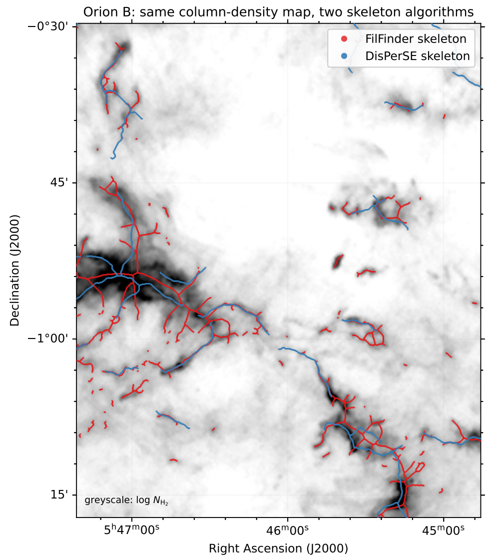

Figure 2. The dominant observational systematic, shown directly: the DisPerSE (blue) and FilFinder (red) skeletons extracted from the *same Herschel* column-density map of Orion B (greyscale, $\log N_{\rm H_2}$; a representative sub-region). The FilFinder mask uses a global threshold of 3×10^{21} cm^{-2} in $N_{\rm H_2}$, an adaptive threshold of 0.1 pc and a branch threshold of 40 pc; the DisPerSE skeleton uses a 3σ persistence cut. The two algorithms trace substantially different filament networks, FilFinder's medial axis of a column-density mask is denser and more branched and follows more low-contrast material, whereas DisPerSE follows the persistent crests, which is why the inferred $s_{\rm ACS,med}/W$ differs by a factor of $\sim$2 between them (Section 2.11). This is intrinsic to how each algorithm defines a "filament" and is the largest single uncertainty on the *magnitude* of the spacing, though the sub-classical *sign* holds for both.

most associated cores lie on coherent crests. We carry it as a one-sided systematic in Table 8. As a floor, retaining the pre-Gaia distances lowers the primary-region characteristic to $s_{\rm ACS,med}/W \approx 1.4$ (still sub-classical), so the sub-classical *sign* does not depend on the distance revision, though its magnitude does.

The measurement convention, above all the skeleton algorithm, dominates the numerical value: the DisPerSE and FilFinder skeletons of the identical Orion B map (Fig. 2) trace substantially different networks, moving the inferred $s_{\rm ACS,med}/W$ by a factor $\sim$2 while the sub-classical *sign* holds for both (Section 2.11).

2.3 Results

The four primary regions are the independent statistical units throughout this paper; individual cores and spacings within a cloud are not independent, and we propagate this nesting in the hierarchical bootstrap (Section 2.11). With only four independent clouds we treat the cloud-to-cloud scatter ($\sigma \approx 0.4$) as a descriptive range, not a formal confidence interval; we do not attach a bootstrap CI to an $N = 4$ sample.

Two statistics, two different answers. Because this distinction governs everything that follows, we state it before any numbers. The adjacent-core spacing $s_{\rm ACS,med}$ is a *conditional* descriptor: it is measured only where cores are adjacent, so it characterises the interior of core-bearing stretches and says nothing about filament that has formed no cores. The core line density L/N is the *unconditional* quantity, counting cores per unit length of the whole skeleton, and it is L/N, not $s_{\rm ACS,med}$, that a one-fragment-per-wavelength theory predicts. The two give different answers, and that difference is the main observational result of this paper. Conditionally, the spacing is sub-classical by a factor $\approx$ 2.4–4.3 (Table 4). Unconditionally, $(L/N)/W \approx 3.4$, 3.4, 4.5 and 5.8 for Orion B, Aquila, Taurus and Perseus, so Perseus is essentially classical, Taurus close to it, and only Orion B and Aquila fall clearly below the nominal 5–6 W (Section 2.8). The reconciliation is that cores occupy only 6–27 per cent of the filament length (Section 2.9). Every comparison of $s_{\rm ACS,med}/W$ with 5–6 W in what follows, including Table 4 and Fig. 4, is therefore a comparison between a conditional local descriptor and a fragmentation wavelength. It is not a measurement of a shortened fragmentation wavelength, and we do not use it as one.

Primary sample.[1] We adopt the four regions Orion B, Aquila, Perseus and Taurus ($N > 500$ cores each, well-established distances) as the primary sample.

Full sample result. The core-count-weighted mean across all 8 HGBS regions (0.279 pc) and the primary-only weighted mean (0.290 pc) differ by $\approx$ 3.9%, and a leave-one-out analysis confirms no single region dominates ($<$ 6% shifts).

All weighted means use core-count weighting,

$$\bar{\lambda} = \frac{\sum_i N_i \lambda_i}{\sum_i N_i}, \tag{1}$$

where λ_i and N_i are the per-region spacing and core count.

The sub-classical result is a property of the prestellar population, not of protostellar contamination: the prestellar-only ACS median is $s_{\rm ACS,med}/W = 2.10$, indistinguishable from

[1] Throughout, *primary* is a sample-selection label ($N > 500$ cores, secure distance, and a skeleton simple enough for unambiguous core-to-filament association; Section 2.5) for the regions retained for the adjacent-core-spacing measurement, and is *not* a claim of statistical robustness.

Table 4. Convention decoder for the *conditional* local descriptor: read every observed $s_{\mathrm{ACS,med}}/W$ quoted in the text against it. This table tabulates the adjacent-core spacing, which is measured only within core-bearing stretches and is *not* the quantity a one-fragment-per-wavelength theory predicts (that is L/N; Section 2.8). Values are given under each width/skeleton convention and correction state for the four primary HGBS regions (revised distances); every combination is conditionally sub-classical (classical $\lambda_{\mathrm{m}} \approx$ 5–6×FWHM). We adopt the directly-measured region-Plummer value as our reference, $\boldsymbol{(s_{\mathrm{ACS,med}}/W)_0 \approx 1.4}$; the fixed-width value (≈ 1.9, retained for comparability with prior HGBS work) and all others are read against it. The "periodic" column applies the periodic injection–recovery correction to the raw value; the data-matched *clustered* column (bracketed) applies ÷1.64. The DisPerSE conventions are sub-classical by a factor $\approx$ 2.4–4.3; across the full envelope the factor spans $\approx$ 2–13, the upper end set by the densest FilFinder skeleton.

Convention	raw	periodic
Region Plummer, DisPerSE (adopted, $(s_{\mathrm{ACS,med}}/W)_0$)	**1.4**	1.2
Fixed 0.10 pc, DisPerSE	2.1	1.9
FilFinder skeleton	0.4–0.8	0.4–0.8
Pairwise median ($L/3$; biased, not a spacing)	2.8	,
Matched simulation (upper limit)	$\lambda/W \lesssim$ 1.0–1.3	
Classical λ_{m} (F&M 2012)	5–6 (= 4× diam.)	

Notes The reference point is the region-Plummer ACS $(s_{\mathrm{ACS,med}}/W)_0 \approx 1.4$ (bold), one illustrative point within the convention range 0.8–1.9; it is an adopted reference value, chosen for comparability with prior HGBS work, not a uniquely preferred physical scale (and not a formal confidence interval; Section 2.11). The data-matched *clustered* branch (÷1.64 on the raw value, giving $\approx$ 0.8 region-Plummer, $\approx$ 1.2 fixed-width and $\approx$ 0.3–0.5 FilFinder) is *not* shown as a column because it is not independent: it applies a correction derived by *assuming* the observed clustering (Section 2.7), so it is circular and serves only as a lower bound; we carry the raw value throughout. The fixed-width convention gives a *higher* value (≈ 1.9) because it uses 0.10 pc rather than the larger measured widths; FilFinder gives lower values still. The full envelope factor $\approx$ 2–13 (Section 2.7) is $\lambda_{\mathrm{m}}/(s_{\mathrm{ACS,med}}/W)$: its lower endpoint ≈ 2 uses $(\lambda_{\mathrm{m}}, s_{\mathrm{ACS,med}}/W) = (5.5W, 2.6)$ (fixed-width DisPerSE, Perseus) and its upper endpoint ≈ 13 uses $(5.5W, 0.42)$ (densest FilFinder skeleton). We quote $s_{\mathrm{ACS,med}}/W$ to one decimal place; finer digits are not significant.

the all-cores 2.08 (Section 2.12). The preferred ACS measurement follows in Section 2.7.

2.4 The dominant observational systematic: filament width

Two width conventions recur throughout: the *fixed-width* convention (the canonical HGBS $W_{\mathrm{fil}} = 0.10$ pc for every region) and the *region-Plummer* convention (a beam-deconvolved Plummer FWHM fitted per region); since $s_{\mathrm{ACS,med}}/W \propto 1/W$, the broader region-specific widths give a *lower* $s_{\mathrm{ACS,med}}/W$. The fixed $W_{\mathrm{fil}} = 0.10$ pc convention is the largest single lever on $s_{\mathrm{ACS,med}}/W$, comparable to the whole discrepancy, so we measured region-specific widths directly. Along each spine we sampled the column-density profile perpendicular to the local tangent and fitted a beam-deconvolved Plummer function (the HGBS width definition; Arzoumanian et al. 2019) to hundreds of profiles per region (18.2″ beam; Table 5). All four regions give $p \simeq 2.0$ and $W_{\mathrm{fil}} = 0.11$–0.16 pc, agreeing with the published HGBS values and slightly broader than 0.10 pc, so the region-specific $s_{\mathrm{ACS,med}}/W$ (1.1–1.8, median ≈ 1.4) is *lower* than the raw fixed-width value ($\approx$ 2.1; Table 4); every region is sub-classical under either convention. A Gaussian-crest estimate returns a spuriously narrow Taurus width (0.031 pc, formally $s_{\mathrm{ACS,med}}/W \rightarrow 4$); the physically-motivated Plummer fit (0.11 pc, $s_{\mathrm{ACS,med}}/W \approx 1.1$) makes Taurus the most sub-classical region. The result is thus not conditional on the width convention. We prefer the Plummer widths on physical grounds: they fit the observed $p \simeq 2$ profile of HGBS filaments and are far less sensitive to the fitting radius than either a single fixed 0.10 pc value or a Gaussian inner-ridge fit.

Table 5. Region-specific, beam-deconvolved filament widths from a direct Plummer fit to perpendicular column-density profiles along the skeletons (the HGBS width definition; Arzoumanian et al. 2019), and the resulting ACS $s_{\mathrm{ACS,med}}/W$. W_{fil} is the beam-deconvolved Plummer FWHM (18.2″ beam), p the Plummer index (a *free* fit parameter, not fixed; it converges to $p \simeq 2.00$ in all four regions, only Taurus differs, at 2.01, consistent with the canonical Plummer-2 profile), N the number of fitted profiles. All four regions give $p \simeq 2$ and sub-classical $s_{\mathrm{ACS,med}}/W$ on the (primary-sample) *median* spacing quoted here; the *mean* adjacent spacing is a factor $\approx$ 1.3–2.2 larger (Section 2.10), so the mean-based ratio is $\approx$ 2.1–4.7, still sub-classical in every region.

Region	W_{fil} (pc)	p	N	ACS (pc)	$s_{\mathrm{ACS,med}}/W$
Taurus	0.110	2.01	295	0.124	1.1
Orion B	0.148	2.00	356	0.207	1.4
Perseus	0.147	2.00	235	0.263	1.8
Aquila	0.161	2.00	446	0.210	1.3
median	0.148	2.00	,	,	1.4

The measured widths (0.11–0.16 pc) agree with the published HGBS values and exceed the canonical 0.10 pc, so region-specific Plummer widths give a *lower* $s_{\mathrm{ACS,med}}/W$ ($\approx$ 1.4, raw) than the raw fixed-width value ($\approx$ 2.1; Table 4): all four regions are sub-classical under either convention. ACS spacings are the full-coverage medians (Section 2.7), measured on the same DisPerSE skeleton as the widths (its deposited crest for the width fit, its bridged form for the spacing); the injection–recovery bias correction lowers each $s_{\mathrm{ACS,med}}/W$ by a further $\sim$10–20%. (Pipeline outputs retained to 2 s.f.; the convention systematic dominates.)

The fit is consistent across the sample: every region returns a Plummer index $p \simeq 2.0$, the value found for HGBS filaments generally (Arzoumanian et al. 2019), and a width within $\sim$ 50% of 0.10 pc (inter-quartile ranges $\sim$ 0.07–0.20 pc), so the sub-classical $s_{\mathrm{ACS,med}}/W$ is a property of all four primary regions, not of one anomalous cloud. The single earlier outlier, a spuriously narrow 0.031 pc Gaussian crest for Taurus, was a profile-selection effect of fitting the inner ridge rather than the broad Plummer wings (not a pixel-scale artefact: the 3″ pixels subtend 0.002 pc at 135 pc); it does not survive the Plummer fit, which gives 0.11 pc and $s_{\mathrm{ACS,med}}/W \approx 1.1$.

2.5 Sample Classification and Distance Robustness

We classify the eight regions (Tables 3 and A1) as **primary** (Orion B, Aquila, Perseus, Taurus) or **limited** by three criteria fixed *a priori* (before the measurement): $N > 500$ cores;

a well-established distance; and a skeleton simple enough for unambiguous core-to-filament association. This selection is cross-checked by the leave-one-out test below. The limited sample fails at least one criterion (Ophiuchus on skeleton complexity, the others on small samples) and are excluded from the primary ACS measurement. The largest distance revisions (Serpens +76%, Aquila +68%) are independently corroborated by VLBI maser parallaxes (validating Gaia DR3 to $< 5\%$; Kounkel et al. 2017; Xu et al. 2019; Ortiz-León et al. 2018, 2017), extinction estimates (Yan et al. 2022; Green et al. 2024) and co-spatiality with Aquila (Zucker et al. 2020); Serpens is *limited* and excluded in any case. The sub-classical conclusion is independent of these regions: leave-one-out shifts the weighted mean by $< 6\%$, and dropping Aquila leaves the region-Plummer median at 1.40 and the fixed-width characteristic at 2.1, both sub-classical. The primary/limited split is likewise not tuned to the distance revision: its criteria act on the distance-independent core count and skeleton morphology, the closest any region lies to the $N > 500$ boundary is Taurus at 536 cores ($\sim 7\%$ above), and because the Gaia revision rescales spacings but not core counts, no plausible distance change moves a region across the classification.

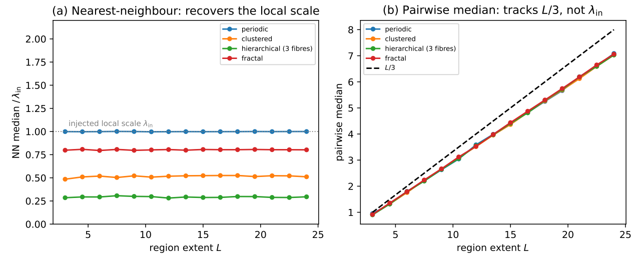

Figure 3. Synthetic validation that the adjacent-core spacing (ACS) statistic recovers the injected local spacing scale while the pairwise median does not (a clustered process need not possess a fragmentation wavelength). Cores are placed on a segment of extent L by four point processes (periodic, clustered, hierarchical three-fibre, fractal), each with a fixed injected local scale $\lambda_{\rm in}$. *(a)* The ACS median (normalised by $\lambda_{\rm in}$) is flat with L and recovers each process's local/apparent scale (the hierarchical case gives $\lambda_{\rm in}/3$, the projected value). *(b)* The pairwise median instead rises along $0.293\,L$ (dashed; this is the median of the triangular pairwise-distance law, whose *mean* is $L/3$; Appendix B) for *all four* processes, independent of $\lambda_{\rm in}$; confirming that it measures the segment extent, not the fragmentation wavelength.

2.6 Choice of spacing statistic: the region-extent bias of the pairwise median

We adopt the along-skeleton adjacent-core spacing (ACS), denoted $s_{\rm ACS,med}$ for its per-region median, as the primary measure of the core spacing, reporting the all-pairs median only for comparability with prior HGBS work. This is *not* the Euclidean nearest-neighbour distance, which we use only for the Clark–Evans statistic in Section 2.10. In figures whose axis reads λ/W for continuity with prior work, the plotted observed quantity is $s_{\rm ACS,med}/W$, a line-density statistic, never a wavelength. The two differ fundamentally: for N points uniformly distributed on a filament of length L the pairwise median converges to $d^\star = L(1-1/\sqrt{2}) \approx 0.293\,L$ for uniform, extent-filling distributions (i.e. $\sim L/3$), the extent of the filament segment ($L_{\rm fil} \sim 1\,$pc), for a random or a perfectly periodic distribution alike, so it measures the extent, not the local core spacing (triangular-distribution derivation, finite-N Monte-Carlo and branched-skeleton generalisation in Appendix B). It tracks $\sim L/3$ for any filament with $N \gtrsim 5$ cores and never approaches the local spacing (~ 30–$100\times$ smaller). The ACS statistic is unaffected by this bias: the $18.2''$ HGBS beam corresponds to 0.012–0.039 pc, $\gtrsim 6\times$ smaller than the measured ACS spacing, but because source extraction and deblending operate on a scale larger than one nominal beam this beam ratio is not by itself sufficient to establish the absence of resolution bias; the relevant argument is the injection–recovery test (Section 2.7), which recovers the input spacings to a few per cent.

We validate this directly with synthetic point processes (Fig. 3). For core distributions built as periodic, clustered (gamma-renewal), hierarchical (three projected fibres) and fractal (power-law-gap) processes with a fixed injected local scale $\lambda_{\rm in}$, the ACS median is *flat* with segment extent L and recovers each process's local scale, whereas the pairwise median rises as $0.29\,L$ for *every* process, independent of $\lambda_{\rm in}$; i.e. it estimates the segment extent, not the fragmentation scale. This confirms the analytic result across the full range of plausible core-placement statistics.

We reconstruct the along-skeleton core ordering for the four primary regions from the deposited DisPerSE skeleton maps, bridging fragmentation by morphological closing, and extend the ACS measurement to all four at full coverage in Section 2.7; the limited sample is excluded from the ACS measurement owing to branching skeletons or small samples. Figure 4 shows the per-region spacings with revised distances.

2.7 Injection–recovery validation and model dependence of the correction

To test whether the ACS statistic recovers known spacings under realistic HGBS conditions, we placed synthetic cores with known periodic spacings (0.15–0.35 pc) on HGBS-property-matched DisPerSE skeletons for all 8 regions. On clean skeletons the raw measurement recovers the input to $3.05 \pm 0.69\%$; the larger correction actually applied is the *full-coverage pipeline* bias ($\div 1.11$–1.29, from junction/branch topology on the real bridged skeletons), which moves raw $2.1 \to 1.9$ and is a genuine, region-measured correction. We do *not* additionally apply a point-process model factor (periodic $\div 1.20$ vs clustered $\div 1.64$): if the cores are clustered the measured ACS median already *is* the physical line-density quantity, so correcting it toward a periodic template the data reject would be circular. We therefore carry the raw value as the adopted $(s_{\rm ACS,med}/W)_0$ and quote the clustered branch ($\div 1.64$: raw $2.0 \to 1.2$) only as a lower bound. The sub-classical *sign* holds under every choice.

Repeating the exercise for the pairwise median confirms it does not recover $\lambda_{\rm true}$ (it returns a value $\sim 20\times$ larger, tracking $L/3$), whereas the ACS statistic recovers $\lambda_{\rm true}$ to 1.8–3% on the identical skeletons (Appendix B).

Full-coverage along-skeleton ACS. We extend the ACS measurement to all four primary regions using the dense deposited DisPerSE skeletons, which (unlike the sparse

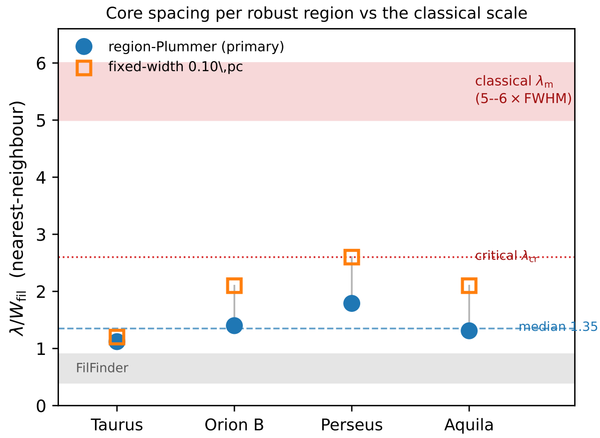

Figure 4. Adjacent-core (ACS) spacing per primary region ($s_{\rm ACS,med}/W_{\rm fil}$), the *conditional* descriptor measured within core-bearing stretches; the unconditional line density L/N, which is what fragmentation theory predicts, is shown in Fig. 5, in the two DisPerSE width conventions: directly-measured region-Plummer widths (blue circles, preferred; per-region raw $1.1, 1.4, 1.8, 1.3$, dashed line at the median, which we quote as 1.4; the rounded median of these four values, whose exact median is 1.35) and the fixed 0.10 pc width (orange squares; $1.2, 2.1, 2.6, 2.1$), both shown *raw* (before the injection–recovery correction; Section 2.7). The red band is the classical fastest-growing wavelength $\lambda_{\rm m} \approx 5$–$6\times$FWHM (Fischera & Martin 2012; $= 4\times$ the flat diameter, Inutsuka & Miyama 1992) and the dotted line the classical *critical* wavelength $\lambda_{\rm cr} \approx 2.6\times$FWHM; every region lies well below both. The grey band marks the independent FilFinder skeleton range ($\lambda/W \approx 0.4$–0.8). (Uncertainty convention: Section 2.11.)

FilFinder spines) pass near essentially all cores. Bridging the fragmented skeletons by morphological closing plus 1-pixel re-skeletonization, ordering cores by graph arclength and associating within a half-width ($\sim$0.06 pc), we obtain $N_{\rm pairs} = 461, 614, 563, 168$ spacings (Taurus, Orion B, Perseus, Aquila), median ACS $0.124, 0.207, 0.263, 0.210$ pc, i.e. $s_{\rm ACS,med}/W = 1.2, 2.1, 2.6, 2.1$ (characteristic ≈ 2.1; robust to the bridging/association radii). The periodic pipeline factor gives $s_{\rm ACS,med}/W \approx 1.0, 1.9, 2.1, 1.8$ (characteristic ≈ 1.9). Applied to every width convention (Section 2.4), the fixed-width characteristic is $s_{\rm ACS,med}/W \approx 1.2$–$1.9$ and the region-Plummer ≈ 0.8–1.4; the primary observed result is this DisPerSE envelope $s_{\rm ACS,med}/W \simeq 0.8$–$1.9$ (wider under other filament definitions; Fig. 9), within which the raw region-Plummer $(s_{\rm ACS,med}/W)_0 \approx 1.4$ is a bookkeeping reference convention (not the observed value) carried with the correction as a uniform systematic. **The full-coverage ACS spacing ($s_{\rm ACS,med}/W \approx 1.2$–$2.1$), which is conditional on being inside a core-bearing stretch, is sub-classical by a factor ≈ 2.4–4.3 across the DisPerSE conventions (≈ 3.6–4.3 for the region-Plummer reference alone; more under FilFinder), while the pairwise-median ratio ≈ 2.8 (in W units) overestimates it by $\sim$45–50% through the $L/3$ bias.** Taurus is the most sub-classical region, not an outlier. The two high-coverage clouds (Perseus, Taurus; 76/89% core–skeleton association) alone give the same median ($s_{\rm ACS,med}/W = 1.8$ and 1.1, median 1.4), so the result is not driven by the low-coverage clouds.

As an independent *segmentation-method* cross-check, it uses the same *Herschel* maps and source catalogue, so it is not an independent dataset, only an independent way of defining the filament network, running the same along-skeleton BFS-arclength ACS pipeline on independently extracted FilFinder spine maps (all four primary regions) gives $s_{\rm ACS,med}/W \approx 0.4/0.6/0.7/0.8$ for Taurus, Orion B, Perseus, Aquila respectively, systematically 2–3$\times$ smaller than the DisPerSE values (1.0/1.9/2.1/1.8) because the FilFinder mask threshold produces a denser, more branched network. Both methods agree the characteristic spacing is far sub-classical ($s_{\rm ACS,med}/W \ll 4$) in every region; FilFinder if anything strengthens the conclusion. The injection–recovery estimator test (Section 2.7) was run on the DisPerSE skeletons; for FilFinder, its denser, more-branched topology (Fig. 2) is precisely why it returns shorter spacings, so we carry the FilFinder value only as the extreme of the convention envelope, not as a load-bearing number; a matched FilFinder injection–recovery, reported in the next paragraph, confirms the shorter values are genuine rather than an estimator artefact.

Model dependence of the correction. Deriving for each region a correction matched to its own measured clustering (anchored to the periodic $\div 1.20$ and Poisson $\div 1.64$ endpoints) gives region-matched factors $\div 1.40$–1.86 and corrected $s_{\rm ACS,med}/W \approx 0.7$–$1.6$ (fixed $W = 0.10$ pc). The effect on the characteristic is small (mean $s_{\rm ACS,med}/W$ $1.2 \rightarrow 1.3$): the sub-classical result holds under each region's own matched correction, and $s_{\rm ACS,med}/W \approx 1.9$ (which requires the rejected periodic model) remains excluded.

As a matched, independent-segmentation test, we ran the same synthetic injection–recovery on the deposited FilFinder spines: cores placed at known arclength spacings (0.15–0.35 pc) are recovered to $< 1\%$ (mean $|{\rm err}| = 0.2$–1.0% across the four clouds). The shorter FilFinder $s_{\rm ACS,med}/W$ is therefore *not* an estimator or branching artefact; it reflects that FilFinder traces a denser, more-branched network; a genuinely different definition of "the filament" (Section 2.14), not a measurement bias.

2.8 Direct line density, the local-width test, and the coverage bias

The median adjacent spacing measures the spacing *where cores are adjacent*; the quantity a one-fragment-per-wavelength theory actually predicts is the core *line density*, cores per unit filament length. We therefore report three complementary statistics (Fig. 6), computed in a single independent re-implementation and all normalised by the *same* fixed 0.10 pc width so that they are directly comparable (their absolute values differ from the adopted pipeline by less than the convention systematic; Section 2.11). (i) The median adjacent (along-skeleton adjacent-core) spacing is $s_{\mathrm{ACS,med}}/W \approx 1.7$–3.0, sub-classical in every cloud; these single-re-implementation fixed-width medians reconcile with the adopted-pipeline fixed-width per-region values (1.2–2.6; Table 4) to within the convention systematic (Taurus differs most, $\sim 40\%$). (ii) The mean adjacent spacing is a factor 1.3–2.2 larger (≈ 2.1–4.7), but on the heavily-bridged skeletons it is inflated by arclength-ordering jumps across branches (the same effect that lifts Perseus's ACS above its pairwise median; Section 2.6), so we treat it as an upper descriptor, not the line density. (iii) The direct line density N/L (associated cores divided by total skeleton arclength) is the ordering-independent inverse-line-density statistic and the correct quantity to set against a one-fragment-per-wavelength prediction: $L/N = 0.34$–0.58 pc, i.e. $(L/N)/W \approx 3.4$ (Orion B), 3.4 (Aquila), 4.5 (Taurus), 5.8 (Perseus). It is a factor ≈ 2 larger than the median adjacent spacing in every cloud. **On the direct line density the deficit is therefore much milder: Perseus is essentially classical and Taurus near-classical, while Orion B and Aquila remain sub-classical at $\approx$ 3.4.** The two contributions to this near-classical global value can be separated, and neither is small-scale clustering. The median→mean rise (≈ 40–120%) is the *generic mean/median skew* of a broad gap distribution (a pure exponential already gives 1.44; Section 2.10), not clustering; the further mean→ L/N rise is *dilution by core-poor filament*; extensive skeleton stretches with no cores (a coverage effect). The line density restricted to core-bearing stretches (the mean adjacent spacing, ≈ 2.1–$4.7\,W$) is intermediate and still mildly sub-classical, so the global near-classical L/N is *not* evidence of a uniform classical spacing: it is an intermittent distribution with sub-classical local spacing and a small-scale exclusion, whose filament-averaged density is lifted toward classical by the skew and by long core-free filament. The median adjacent ratio adopted elsewhere is accordingly a *descriptor* of the local spacing within core-bearing stretches; L/N (core-bearing or full) is the quantity to compare with a one-fragment-per-λ_{m} theory.

Two further tests support this reading. First, a *local-width* normalisation (a genuinely like-for-like test): for every adjacent-core interval we fit the beam-deconvolved transverse width at the interval midpoint and form the per-cloud distribution of $\Delta s/W_{\mathrm{local}}$ (146–545 reliable fits per cloud). The medians are $0.61, 0.91, 1.05, 0.99$ (Taurus, Orion B, Perseus, Aquila), *all* sub-classical, with only a weak spacing–width correlation ($r = -0.06$ to 0.31). These median-of-ratios values are if anything *more* sub-classical than the corresponding ratio-of-medians (≈ 1.4–1.9 in the skewed clouds), differing by up to a factor ~ 2 in Taurus and Perseus; the per-segment normalisation therefore *strengthens* the sub-classical conclusion rather than depending on the two agreeing. Second, the along-skeleton *pair-correlation* function $g(s)$ (Fig. 7) shows *no* peak at the classical scale 5–$6\,W$ (≈ 0.5–0.8 pc): $g \approx 1$ there (slightly below unity in Orion B and Aquila), with a mild small-separation excess in Taurus and weak region-envelope features near 1 pc. There is thus no classical-scale peak detected; within the $N = 4$ noise the classical periodic comb is not present.

Finally we quantify the skeleton-coverage bias. Only a fraction of catalogue cores associate with the primary skeleton (24–89% across the four clouds; Table 9), and the omission is not neutral: the rejected cores have a 2-D projected core-to-core spacing 1.5–3.0× *larger* than the associated cores in every cloud, so including them would lengthen the measured spacing. Coverage selection therefore biases the adjacent spacing *short*, in the direction of the sub-classical result. We now *bound* it: relaxing the association half-width from 0.06 to 0.30 pc adds the rejected cores back, raising the coverage to 70–97%, and lengthens the median adjacent spacing by only +3% (Taurus), +8% (Perseus), +11% (Orion B) and +16% (Aquila); it stays sub-classical throughout. The coverage bias is thus one-sided and bounded at $\lesssim 16\%$; we carry this magnitude as an explicit systematic (Table 8). (The coverage fractions, three-tier statistics and this bound come from the independent re-implementation at its own association radius; they differ from the adopted-pipeline values by less than the convention systematic, and the $\lesssim 16\%$ bound is insensitive to the radius choice.) A cleaner per-segment analysis (splitting each skeleton at junctions and measuring unbranched segments) was attempted but the morphologically-bridged skeletons over-fragment at spurious junctions, so a physically meaningful segmentation would itself require collinear-segment merging, the same "what is a filament" ambiguity discussed in Section 2.14. We defer it; the coverage bound above and the three-tier decomposition address the underlying concern that the graph representation drives the short spacing.

2.9 Intermittency measured: the fragmentation duty cycle

The local-versus-global contrast invites an obvious question: do cores actually occupy preferential stretches of filament? This can be measured rather than inferred, and doing so turns intermittency from an interpretation into an observable. We assign each associated core an occupancy interval of ± 0.06 pc along its spine (one association half-width), merge overlapping intervals, and define the *duty cycle* as the fraction of total skeleton length that is core-bearing. We also count cores in 1 pc cells along the spines, from which the Fano factor $F = \sigma^2/\mu$ measures over-dispersion relative to a Poisson process ($F = 1$), and compute the Gini coefficient of the same counts.

Table 6 collects the results, and Fig. 5 displays them. The duty cycle is small in every cloud: between 6 and 27 per cent of the skeleton carries cores, so most filament length is empty at HGBS sensitivity. The Fano factor exceeds unity everywhere (1.6, 2.8, 5.7, 25.2 for Aquila, Orion B, Perseus and Taurus), which is significant over-dispersion on parsec scales, and the Gini coefficients (0.36–0.57) confirm that cores are distributed very unevenly along the spines. Within the core-bearing components the core-free runs are short (median 0.05–0.11 pc, 90th percentile 0.24–0.51 pc), but they account for only part of the empty length: in Orion B, for example, 92 pc is occupied and a further 155 pc lies in core-free runs inside core-bearing components, leaving roughly 90 pc of the

Table 6. Intermittency diagnostics. $L_{\rm tot}$ is the total skeleton length, $L_{\rm core}$ the core-bearing length (union of ± 0.06 pc intervals about each associated core), and the duty cycle is their ratio. $f_{\rm assoc}$ is the fraction of catalogue cores associated with the skeleton, F the Fano factor of counts in 1 pc cells and G their Gini coefficient. Both association half-widths are listed. Values come from the independent re-implementation (Section 2.8).

Region	half (pc)	$L_{\rm tot}$ (pc)	$L_{\rm core}$ (pc)	duty	$f_{\rm assoc}$	$\frac{L/N}{W}$	$\frac{L_{\rm core}/N}{W}$	F	G
Orion B	0.06	337	92	0.27	0.53	3.4	0.9	2.8	0.49
	0.30	337	117	0.35	0.85	2.1	0.7	4.5	0.50
Aquila	0.06	106	29	0.27	0.41	3.4	0.9	1.6	0.40
	0.30	106	38	0.36	0.70	2.0	0.7	2.1	0.36
Perseus	0.06	290	35	0.12	0.61	5.8	0.7	5.7	0.43
	0.30	290	40	0.14	0.85	4.2	0.6	10.5	0.47
Taurus	0.06	210	13	0.06	0.86	4.5	0.3	25.2	0.57
	0.30	210	14	0.07	0.97	4.0	0.3	27.3	0.57

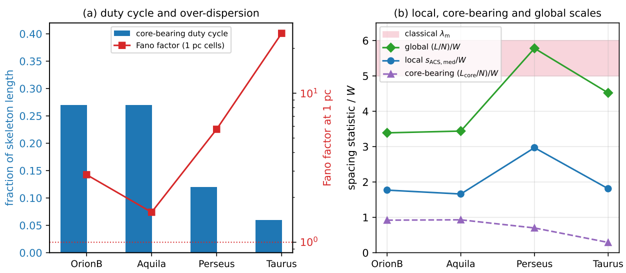

Figure 5. Intermittency of core formation along the filaments. *(a)* The core-bearing duty cycle (bars, left axis) is 6–27 per cent, while the Fano factor of counts in 1 pc cells (red, right axis, logarithmic) exceeds the Poisson value of unity in every cloud. *(b)* The three normalised spacing statistics: the local adjacent-core spacing is well below the classical band, the global $(L/N)/W$ approaches it, and the core-bearing $(L_{\rm core}/N)/W$ lies below both because the occupied stretches are densely populated. The separation between the blue and green curves is the intermittency.

337 pc total in components containing no catalogued core at all.

Two length scales therefore coexist. On sub-parsec scales the gap distribution shows a deficit of the smallest separations (Section 2.10), which is mild regularity rather than clustering. On parsec scales the counts are strongly over-dispersed. That combination is the signature of an intermittent process: cores form in groups with short internal separations, and the groups are separated by long stretches that have produced none. It also explains the factor of about two between the local and global statistics without appealing to small-scale clustering, since the filament-averaged line density is diluted by filament that has not fragmented. The quantity a fragmentation model must reproduce is thus not only a spacing but a spatial duty cycle: where along a filament fragmentation happens, as well as how far apart the fragments lie.

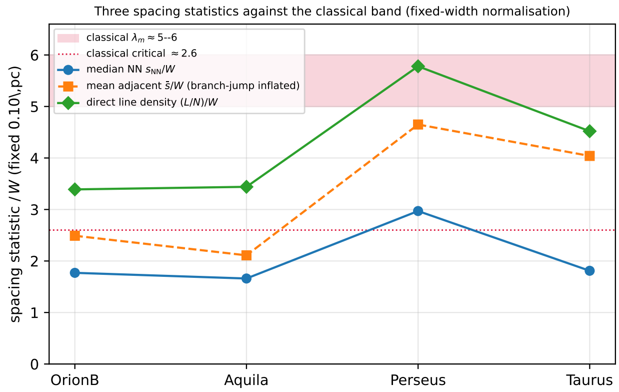

Figure 6. Three spacing statistics per cloud (single re-implementation, all normalised by the fixed 0.10 pc width so they are directly comparable), against the classical band (5–6, red; critical ≈ 2.6, dotted). The median along-skeleton ACS spacing (blue, ≈ 1.7–3.0) is sub-classical everywhere; the direct line density $(L/N)/W$ (green, ≈ 3.4–5.8) is classical in Perseus, near-classical in Taurus, and sub-classical in Orion B and Aquila; the mean adjacent spacing (orange) lies between but is inflated by arclength branch-jumps. The factor-≈ 2 gap between the blue and green points reflects the generic mean/median skew plus dilution by core-poor filament (a coverage effect), not small-scale clustering (Section 2.8): short adjacent spacing within core-bearing stretches, diluted by extensive core-free filament.

Figure 7. Along-skeleton pair-correlation function $g(s)$ for the four primary clouds (uniform-random expectation on the matched components divided out; $g = 1$ is no correlation). Dotted vertical lines mark the classical scale $5.5\,W_{\rm fil}$ per region. No cloud shows a peak at the classical scale ($g \approx 1$ there, slightly below unity in Orion B and Aquila); the only departures are a mild small-separation excess (Taurus) and weak region-envelope features near 1 pc. The classical periodic comb is therefore absent, consistent with a non-periodic, intermittent distribution bearing a small-scale exclusion (Section 2.10).

2.10 Regularity of the core spacing: periodic, random, or clustered?

Classical fragmentation predicts a *quasi-periodic* chain of cores; fibre projection or a turbulent/accretion-driven cascade predicts a broad, clustered distribution with no single comb. Ordering associated cores by arclength along each spine and measuring adjacent-core gaps, we find a coefficient of variation CoV $= \sigma/$mean of order unity (Taurus 1.36, Orion B 0.83, Perseus 0.96, Aquila 0.65; pooled 1.16; Fig. 8).

We calibrated the goodness-of-fit tests by parametric bootstrap, resampling from the fitted distribution to build the exact null distribution of the Kolmogorov–Smirnov statistic and so removing the estimated-parameter bias of the standard test. The resampled gaps are drawn as *independent renewal intervals* from the fitted exponential, not as gaps rederived from a resampled point pattern. The result is two-sided and unambiguous: the regular "periodic+jitter" model is rejected overwhelmingly ($p_{\rm boot} < 10^{-5}$ in every region, at the parametric-bootstrap floor $1/N_{\rm boot}$ with $N_{\rm boot} = 10^5$), *and* the pure-exponential (Poisson) model is also rejected ($p_{\rm boot} < 10^{-5}$ in Orion B, Aquila and Taurus, and 1.7×10^{-3} in Perseus). The cores are therefore neither periodic nor pure-Poisson. The exponential fails because of a deficit of the smallest separations (1-D Clark–Evans ratio $R > 1$), a finite-core/deblending exclusion at the $\sim$beam scale. Comparing five renewal gap models by the corrected Akaike information criterion (AICc; Table 7) makes the two-sided result quantitative and robust to the model set: the pure-Poisson (exponential) model is disfavoured (ΔAICc= 37–246) and the periodic (Gaussian-comb) model strongly rejected (ΔAICc= 81–742) in every region, so the cores are neither periodic nor pure-Poisson. The specific hard-core (shifted-exponential) model is *not* uniquely preferred once broader alternatives are admitted: a positively-skewed lognormal fits best in three of the four regions (the shifted-exponential is best only in Perseus, where lognormal is ΔAICc= 2.5 behind). The data therefore favour a broad, skewed distribution truncated at small gaps rather than a sharp physical hard core. The inferred exclusion scale is 0.011–0.044 pc, of order the beam/core diameter; we therefore treat it as an *observational exclusion scale* with an instrumental component (beam and deblending) plus a possibly-physical component that cannot be separated at this resolution, rather than as a demonstrated physical hard core. Because the many gaps in a structured cloud are not independent draws, we treat these $p_{\rm boot}$ values and AICc differences as *conditional* goodness-of-fit statistics at fixed skeleton, not population-level significance tests; the four clouds remain the independent astrophysical units (Section 6.4).

The cores are consistent with a point process bearing a beam-scale exclusion (largely instrumental, and driving the Poisson rejection) plus region-dependent tail clustering (super-Poisson in Taurus, sub-Poisson in Aquila and Orion B, so "clustered" is a population-level description). There is no single characteristic comb, the observational counterpart of the Clarke et al. (2020) result that fragmentation through sub-filaments leaves no single characteristic length.

Two consequences follow. First, because the gap distribution is broad and positively skewed the mean adjacent spacing exceeds the median by a factor $\approx$ 1.3–2.2 (Orion B 1.4, Aquila 1.3, Perseus 1.6, Taurus 2.2; 1.44 for a pure exponential); but neither the mean nor the median gap is the inverse

Table 7. Gap-model comparison by the corrected Akaike information criterion (AICc; relative to the best model, ΔAICc= 0), with the number of free parameters k. Five renewal models are compared: Poisson (pure exponential), shifted-exponential (hard-core), periodic (Gaussian comb), gamma and lognormal. Both the periodic (ΔAICc= 81–742) and the pure-Poisson (ΔAICc= 37–246) models are strongly disfavoured in every region, so the two-sided "reject both periodic and Poisson" result is robust to the model set; but the specific hard-core shifted-exponential is *not* uniquely preferred, a broad, positively-skewed lognormal fits best in three of the four regions, so the small-gap deficit is better read as an exclusion-limited (largely instrumental) truncation of a broad distribution than as a demonstrated physical hard core. AICc corrections are negligible here ($n \gg k$). This comparison is *conditional* on the fixed deposited skeleton and is a per-region, not a population-level, statement; with only $N = 4$ clouds it does not establish a population-level gap law.

Region	Poisson (k=1)	shifted-exp (k=2)	periodic (k=2)	gamma (k=2)	lognormal (k=2)
Orion B	245.8	46.4	580.0	87.1	**0**
Aquila	82.8	5.8	81.1	11.0	**0**
Perseus	37.1	**0**	367.3	22.4	2.5
Taurus	98.3	76.1	741.8	89.9	**0**

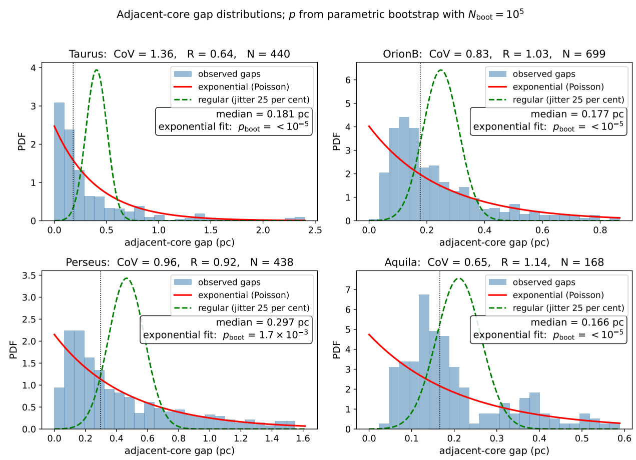

Figure 8. Adjacent-core gap distributions along the skeletons of the four primary regions (histograms), with exponential/Poisson (solid) and regular/periodic (dashed) reference curves; each panel gives the coefficient of variation CoV, the 1-D Clark–Evans ratio R, and the exponential goodness-of-fit p-value *calibrated by parametric bootstrap* ($p_{\rm boot}$; the null distribution of the KS statistic is built by resampling from the fitted exponential, removing the estimated-parameter bias of the standard test). The result is two-sided: the regular/periodic model is rejected overwhelmingly ($p_{\rm boot} < 10^{-5}$, the parametric-bootstrap floor, $1/N_{\rm boot}$ with $N_{\rm boot} = 10^5$) in every region, *and* the pure-exponential (Poisson) model is also rejected ($p_{\rm boot} < 10^{-5}$ (Perseus 1.7×10^{-3}) in all four), so the cores are neither periodic nor pure-Poisson; consistent with a clustered process bearing a beam-scale exclusion (the deficit of small gaps, $R > 1$; largely instrumental).

line density, which is the ordering-independent N/L over the whole skeleton (Section 2.8), and it is N/L that should be set against a one-core-per-wavelength model. We retain the median as the robust descriptor of the observed distribution (Table 5). Second, this fixes the logical status of the comparison: a fragmentation process still sets a characteristic *mean* separation even when the point pattern is not a sharp comb, which clustering, the beam-scale exclusion and projection broaden into the observed near-exponential distribution. The *intra-group* adjacent spacing (median ACS) is sub-classical whereas the filament-averaged line density N/L is near-classical (Taurus, Perseus) and sub-classical only in Orion B and Aquila; there is thus no globally large excess of cores per unit length, and we read all spacing-vs-wavelength comparisons below as order-of-magnitude consistency of a characteristic scale, not a match of a resonant wavelength.

2.11 Combined Systematic Error Budget

We separate three kinds of term (Table 8): *statistical/sensitivity* terms (the cloud-to-cloud and choice-dependent scatter that would change under resampling), *systematic* terms (deterministic offsets from a fixed methodological choice, e.g. the pipeline bias correction), and *modelling* terms (arising from the simulation set-up, e.g. the line-mass dependence). The uncorrelated statistical/sensitivity terms are combined in quadrature,

$$\sigma^2_{\rm sys} = \sum_i \sigma_i^2, \tag{2}$$

and the multiplicative width- and skeleton-convention factors are then applied *afterwards* as separate *analysis-choice envelopes*, not added into this quadrature sum. The adopted region-Plummer ACS result $(s_{\rm ACS,med}/W)_0 \approx 1.4$ (raw fixed-width analogue ≈ 2.1) carries residual pipeline bias ($\sim$10%), protostellar migration ($\sim$7.5%) and between-region scatter ($\sim$4%) in quadrature, plus a $\lesssim 5\%$ manual-spur proxy (Section 2.2); the fixed-width ($\sim$25%; Panopoulou et al. 2022, counted once) and skeleton (factor $\sim$2) convention terms are the analysis-choice envelopes applied on top. The skeleton algorithm is the single largest observational term (Fig. 2; discussed below). Because the dominant term is a deterministic offset we treat the aggregate as a min–max envelope, $s_{\rm ACS,med}/W \approx 0.8$–$1.9$ (width-convention min–max, DisPerSE; Table 4), rather than a Gaussian σ (the skeleton factor of two is carried separately). The adopted headline is $(s_{\rm ACS,med}/W)_0 \approx 1.4$; the fixed-width $s_{\rm ACS,med}/W \approx 1.9$ is a comparator anchored to that convention, not a competing headline.

Projection. The observed separations are plane-of-sky projections of intrinsically 3-D spacings. For an isotropic distribution of filament orientations the projection factor $s_{\rm obs}/s_{\rm true} = \sin i$ has median 0.87 and mean 0.79 (Monte-Carlo, 2×10^6 orientations; 16–84% range 0.54–0.99), so deprojection would *raise* the intrinsic spacing by $\approx$ 15–27%, moving $s_{\rm ACS,med}/W$ towards the classical value (e.g. the adopted 1.4 becomes 1.6–1.8) and reducing, but not removing, the sub-classical deficit. We do not apply this correction, since it depends on the unknown orientation distribution, but carry it as a one-sided comparison systematic that acts *against* the sub-classical result rather than for it.

The legacy pairwise-median value (≈ 2.8 in W units) is an $L/3$ scale, not a local core spacing (Section 2.6); its bootstrap scatter is $\pm$4.2%, and we quote no σ-level tension with theory for it.

Skeleton-algorithm dependence. The DisPerSE and FilFinder values ($s_{\rm ACS,med}/W \approx 1.9$ and ≈ 0.4–0.8; Fig. 2) bracket a genuine ambiguity in what constitutes "the filament"; both lie far below $\lambda_{\rm m}$, so the factor of two moves the magnitude but not the sign. A partial matched-threshold FilFinder extraction shows the inferred spacing is itself strongly sensitive to the mask threshold, so the factor is real and, if anything, a lower bound; an agreed, algorithm-independent segment definition is the important future work. The skeletoniser also selects the interpretation: in matched Plummer units the simulated single-filament upper bound ($\approx$ 1.0–1.3) overlaps the region-Plummer values within the methodological envelope, so a denser FilFinder extraction would need the extra compression of hierarchical-fibre projection (Section 6.2) while DisPerSE would not.

Note on uncertainties: with only four clouds, all σ and $\pm$ values in this paper (including in Table 8 and Fig. 4) are descriptive half-widths of an $N = 4$ range or min–max envelopes, not formal confidence intervals.

The comparison-only terms, the T_1 width normalisation ($\eta_W = 0.59$–0.78; Section 3.2) and the Gaussian-vs-Plummer FWHM factor, enter only when the simulated $\lambda/W_{\rm core}$ is converted to observable units; they do not affect the ACS observational measurement, which is already in $W_{\rm fil}$ units. The $T_1 = 0.61$ (band 0.59–0.78; $\sim$ 30%) normalisation is propagated as a multiplicative systematic on the *simulated* $s_{\rm ACS,med}/W$ alone (via the $W_{\rm core}/[T_1 W_{\rm form}]$ conversion of Eq. 6): the $\pm$0.10 maps to a $\sim$ 30% band on the simulated value when it is set against the region-Plummer baseline, and is carried in the comparison group here rather than on the observed spacing.

The empirical *per-region range* of the bias-corrected fixed-width values is $s_{\rm ACS,med}/W = 1.0$–2.1 (the real cloud-to-cloud spread), and the *methodological envelope* $s_{\rm ACS,med}/W \approx 1.1$–$2.4$ (per-region sweep incl. FilFinder; $\sigma_{\rm sys} \approx 28\%$) is obtained by varying the width, skeleton and association-radius choices; the descriptor "$\sigma \approx 0.4$" is only the half-width of the per-region range.

2.12 ACS pipeline reproducibility, population splits and hierarchical statistics

The along-skeleton ACS pipeline thresholds the deposited DisPerSE skeleton, bridges fragmentation by morphological closing and re-skeletonisation, associates each core to the nearest spine within 0.06 pc, orders cores along each component by graph arclength (double-BFS diameter), and takes adjacent-core separations, rejecting cores not on a primary segment. It reproduces the deposited catalogue counts and raw medians exactly (Table 9). As an internal check against coding error the estimator was independently re-implemented from scratch by the author (a second implementation, *not* an external reproduction), which reproduces the same counts and medians.

The fraction of catalogue cores associating to a primary skeleton segment varies across regions, Orion B 37%, Aquila 24%, Perseus 76%, Taurus 89%, so in Orion B and Aquila the ACS spacing is measured on a minority of cores, the distributed clouds having many cores off any primary crest. Any

Table 8. Summary of the terms on the λ/W comparison, grouped as *observational* (on $s_{\rm ACS,med}$), *simulation* (on $\lambda_{\rm turn}$) and *comparison* (terms that enter only when the two are matched). Statistical/sensitivity terms are combined in quadrature (Eq. 2); the width- and skeleton-convention factors are applied afterwards as a separate multiplicative *analysis-choice envelope*, not as Gaussian uncertainties. All $\pm$ values and the $\sigma \approx 0.4$ cloud-to-cloud scatter are descriptive $N = 4$ envelopes (half-widths or min–max ranges), *not* Gaussian confidence intervals (Section 2.11).

Source	Type	Magnitude
Observational (on $s_{\rm ACS,med}$)		
Bootstrap / cloud-to-cloud	sensitivity	$\sigma \approx 0.4$ ($\sim$20%)
ACS pipeline bias	systematic	$\sim$10% (corrected)
Protostellar migration	sensitivity	$\sim$7.5%
Projection (deprojection)	systematic (one-sided, *raises s*)	+15–27%
LOS blending (Aquila)	systematic (one-sided, *shortens s*)	$\lesssim$ 50% worst case
Skeleton-coverage selection	systematic (one-sided, *shortens s*)	$\lesssim$ 16% (bounded)
Width convention, fixed vs region	analysis-choice env.	$\sim$25%
Skeleton algorithm, DisPerSE/FilFinder	analysis-choice env.	factor $\sim$2
Manual skeleton editing	systematic (proxy)	$\lesssim$5%
Simulation (on $\lambda_{\rm turn}$)		
Growth-rate resolution/epoch	modelling	$\lesssim$5% (converged)
Turbulence sensitivity	sensitivity	<5% (single realisation)
Line-mass (f) dependence	modelling	$\sim$25%
Comparison (matching the two)		
Gaussian vs Plummer FWHM	systematic	factor $\sim$1.5
T_1 width normalisation (η_W)	systematic	$\sim$30%

Table 9. ACS pipeline counts: skeleton components, catalogue cores, cores associated to a skeleton, and adjacent ACS pairs, per primary region.

Region	comps	catalogue	associated	ACS pairs
Orion B	67	1844	680	614
Aquila	14	749	182	168
Perseus	54	816	619	563
Taurus	32	536	479	461

bias runs *toward* the sub-classical result (rejected cores lie on fainter segments with *longer* spacings; Section 2.8), and the result does not depend on the low-coverage regions: restricting to the high-coverage clouds (Perseus, Taurus) gives $s_{\rm ACS,med}/W = 1.8$ and 1.1, median 1.4, identical to the four-region median. It is also stable to the association radius: sweeping the half-width over 0.04–0.08 pc changes the per-region ACS median by only 4–7% (Taurus 6.6%, Orion B 7.3%, Perseus 7.4%, Aquila 4.2%), so the effect is not load-bearing. The complementary *relaxed*-radius coverage test reinforces this: widening the association half-width from 0.06 to 0.30 pc raises the coverage to 70–97% yet lengthens the median adjacent spacing by only +3 to +16% (Section 2.8), so the omitted cores bias the result short by a bounded, one-sided $\lesssim$ 16% and the spacing stays sub-classical.

Splitting the sample by the published classification flags, the prestellar-only median $s_{\rm ACS,med}/W = 2.10$ (717 pairs) is indistinguishable from the all-cores 2.08 (1806 pairs; both $\approx$ 1.9 after bias correction). Prestellar cores dominate the on-skeleton budget in every region and set the result; unbound starless cores give a larger, noisier $s_{\rm ACS,med}/W \approx 3.3$ and protostellar cores are too few (6–80 pairs) to constrain it. The sub-classical result is therefore not an artefact of population mixing.

Hierarchical statistics and angular-resolution robustness.
Cores within a filament are not independent, so we assessed the cloud→filament→core nesting with a three-level nonparametric bootstrap (2000 iterations, carrying spacings that share a central core together; the second level samples the 67, 14, 54, 32 filament components per cloud, Table 9). On a 2-D projected ACS metric (which underestimates the arc length, $s_{\rm ACS,med}/W \approx 1.1$–1.5) the bootstrap median is stable, the dominant uncertainty being the cloud-to-cloud scatter $\sigma \approx 0.4$ (range ~ 0.8 in Taurus to ~ 1.7 in Aquila); core-count and equal-cloud weighting agree to ~ 0.1. We therefore quote this scatter, not a per-pair error, as the relevant uncertainty (Section 2.11).

Angular-resolution blending is negligible: merging cores within one beam shifts the characteristic $s_{\rm ACS,med}/W$ by ≤ 0.004, so distance-dependent blending does not bias the adopted value (Table 10).

2.13 Comparison with Literature

Our revised-distance ACS measurements (Table 11) differ from the original HGBS values by the region-dependent distance revisions; the core-count-weighted primary-region mean rises only $\sim$ 10%, dominated by Aquila.

We decomposed this shift on the real Orion B data by toggling each methodological choice at fixed distance (Table 12). The *statistic definition* dominates: switching from ACS to the pairwise median inflates the spacing by $\approx$ 1.5 (0.207 $\to$ 0.313 pc), accounting for essentially the entire 0.212 $\to$ 0.313 pc difference in Table 11; the remaining choices (skeleton re-extraction, association half-width, 2-D→arclength ordering, and the −3.5% distance revision) are each $\leq$ 15% and largely cancel (Table 12). On our *primary* ACS statistic Orion B is 0.207 pc, within 3% of Könyves et al. (2020), so the apparent shift is a property of the pairwise-median comparability statistic, not the physical spacing; the

Table 10. Summary of validation tests performed in this paper.

Test	What it checks	Result
Injection–recovery (ACS)	ACS bias on real skeletons	3%; pipeline 15–20%
Injection–recovery (pairwise)	$L/3$ convergence	tracks $L/3$, not λ
Resolution convergence	$t_{\rm frag}(128^3)$ vs 256^3	ratio 0.915 ± 0.012
Truelove criterion	cells per $\lambda_{\rm J}$ at measurement	long. 5–7; perp. 1–2
IC sensitivity	Gaussian vs uniform profile	identical
EOS sensitivity	$\gamma = 0.8$–$5/3$	$\gamma < 1$ accelerates; 5/3 suppresses
Region resampling (descriptive)	spread over 4 clouds	0.261–0.298 pc
Leave-one-out	single-region dominance	$< 6\%$ shift
Population splits	prestellar vs all cores	$s_{\rm ACS,med}/W =$ 2.10 vs 2.08
Hierarchical bootstrap	cloud→filament→core	$\sigma_{\rm cloud} \approx 0.4$
Mock-blending	distance-dependent beam	$\Delta s_{\rm ACS,med}/W \leq 0.004$
Region-specific Plummer widths	beam-deconvolved $W_{\rm fil}$	0.11–0.16 pc; $s_{\rm ACS,med}/W =$ 1.1–1.8

ACS re-analysis *shortens* Taurus and *lengthens* Perseus, if anything strengthening the conclusion.

2.14 What constitutes the theoretical filament?

Before comparing with theory we note that the factor-of-two DisPerSE-versus-FilFinder difference (Fig. 2) is not measurement noise but reflects that the two algorithms *define different objects* (persistent crests versus the medial axis of a masked skeleton), so there is no single, algorithm-independent "filament" whose spacing theory should reproduce. Our robustness claim applies to the *sign* of the deficit, every convention gives a sub-classical $s_{\rm ACS,med}/W$, while its *magnitude* cannot be assigned a unique value: the same data yield $s_{\rm ACS,med}/W \approx 1.0$ (region-Plummer FilFinder), ≈ 1.4 (region-Plummer DisPerSE) or ≈ 1.9 (fixed-width DisPerSE). Because segmentation alone moves the answer by as much as the differences between the physical models compared in Section 5, the theory should not be read as discriminating between these values. Figure 9 makes the resulting envelope explicit.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s, the critical line mass for gravitational instability is (Ostriker 1964):

$$\mu_{\rm crit} = \frac{2c_s^2}{G} \tag{3}$$

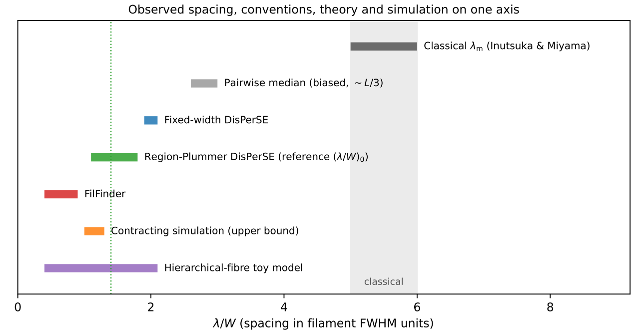

Figure 9. The full methodological envelope of λ/W on a single axis, showing that several distinct physical interpretations occupy the allowed observational region while all lie well below the classical value. Marked, from top: the classical band (5–6; Inutsuka & Miyama 1992; Fischera & Martin 2012); the biased pairwise median ($\approx L/3$); fixed-width DisPerSE (1.9–2.1); region-Plummer DisPerSE (our reference ACS-spacing ratio $(s_{\rm ACS,med}/W)_0 \approx 1.4$); FilFinder (0.4–0.8, Table 4); the contracting-simulation *upper bound* (1.0–1.3); and the hierarchical-fibre toy-model range. Because the observational conventions, the simulated upper bound and the fibre toy model all overlap well below the classical band, the sub-classical *sign* is robust while its *magnitude* is convention-dominated (Section 2.14); the comparison with simulation (Section 5) should be read against this envelope, not against a single number.

The most unstable fragmentation wavelength is, as an *exact* result for the isothermal equilibrium cylinder (Inutsuka & Miyama 1992),

$$\lambda_{\rm IM92} = 22\,H = 4 \times (2R_{\rm flat}), \tag{4}$$

where $H = c_s\,(4\pi G\rho_c)^{-1/2}$ is the thermal scale height, $R_{\rm flat} = \sqrt{8}\,H$ the flattening radius, and $W_{\rm fil} \approx 0.1\,{\rm pc}$ the characteristic filament FWHM. Solving the dispersion relation exactly (Appendix C; Fischera & Martin 2012) gives a fastest-growing wavelength $\lambda_{\rm m} \approx 5 \times W_{\rm fil}$ near-critical, rising to $\approx 6 \times W_{\rm fil}$ at low line mass, and a critical wavelength $\lambda_{\rm cr} \approx 2.6 \times W_{\rm fil}$. The often-quoted "$\lambda \approx 4\times$ the width" is $4\times$ the flat *diameter* $2R_{\rm flat}$ (Inutsuka & Miyama 1992), which equals ≈ 5–$6\times$ the *FWHM*; taking it as $4\times$ the FWHM understates the classical scale. We therefore adopt $\lambda_{\rm m} \approx 5$–$6\,W_{\rm fil}$ (Fischera & Martin 2012) as the benchmark. Because we compare in FWHM units and forward-model the simulations to a beam-convolved Plummer FWHM (Section 3.2), the sub-classical deficit does not depend on the equilibrium-vs-observed profile-index difference ($p = 4$ vs $p \simeq 2$), and we compare as the density-independent ratio $\lambda/W_{\rm fil}$. The deficit is not new in sign: Fischera & Martin (2012) noted observed cores sit "about the FWHM" apart, and Zhang et al. (2020) find a spacing $\approx 1.3\times$ the width in California.

With $f = \mu_{\rm line}/\mu_{\rm crit}$ and $t_{\rm frag} \propto (f-1)^{-1/2}$, marginally supercritical filaments fragment slowly. At magnetically supercritical field strength ($\beta \gtrsim 0.2$) all configurations fragment for $\mathcal{M} \geq 1$ at every f tested once integrated adequately: above the critical line mass there is no *thermal* stability boundary. This is distinct from *magnetic* stabilisation, where a strong longitudinal field ($\beta \lesssim 0.15$) suppresses collapse even for $f > 1$ (the "strong-field, magnetically supported" regime;

Table 11. Comparison with Published HGBS Spacing Measurements

Region	ACS (pc)	Orig. HGBS	pairwise (pc)	Lit. (pc)	Lit. statistic	Reference
Primary sample						
Orion B	**0.207**	0.212	0.313	0.20–0.30	local	Könyves et al. (2020)
Aquila	**0.210**	0.206	0.346	0.22–0.26	local	Könyves et al. (2015)
Perseus	**0.263**	0.199	0.248	0.22	local	Pezzuto et al. (2021)
Taurus	**0.124**	0.206	0.198	0.19	along-fibre	Hacar et al. (2013)
Limited sample						
Ophiuchus	,	0.197	0.206	0.18	local	Könyves et al. (2020)
Serpens	,	0.188	0.331	0.17–0.19	local	Könyves et al. (2020)
TMC1	,	0.203	0.195	∼0.20	along-fibre	Hacar et al. (2013)
CRA	,	0.212	0.248	∼0.21	local	Könyves et al. (2020)

Notes (0) The "Lit. statistic" column records what kind of statistic each cited value represents. All are *local* separation measures, either nearest-neighbour or along-fibre adjacent separations, rather than all-pairs medians, and that is precisely why they agree with our ACS column rather than with our pairwise column; the exact definitions differ in detail between papers. The region-extent bias identified here therefore applies wherever an all-pairs or pooled separation statistic is adopted, including the comparability statistic carried in this paper and in cross-region compilations, and not to these particular published values. (1) The primary statistic is the *adjacent-core* (ACS) spacing (bold); the "pairwise" column is a legacy comparator only, inflated towards $L/3$ (Section 2.6) and not a physical spacing. ACS spacings for the limited sample are omitted. Perseus is the one region where the ACS (0.263) slightly exceeds the pairwise median, because arclength ordering on its heavily bridged skeleton inflates path distances; we verified this is not a bridging artefact, an independent minimum-spanning-tree ordering of the same cores gives a consistent value, insensitive to the morphological bridging, and both remain sub-classical. (2) "Original HGBS" values use pre-Gaia distances; literature ranges reflect different methods and distance assumptions. (3) For Orion B the pre-Gaia distance was already ∼400 pc (Könyves et al. 2020), so the pairwise 0.212 → 0.313 pc difference is a statistic-definition effect, not a distance revision.

Table 12. Decomposition of the apparent Orion B spacing shift (0.212 → 0.313 pc in Table 11) into its methodological contributions, toggled one at a time at fixed distance. The statistic definition (ACS→pairwise median) dominates; the remaining choices are small and largely cancel.

Contribution	Effect on spacing
Statistic definition (ACS → pairwise median)	×1.51 (0.207→0.313 pc)
Skeleton re-extraction	$\lesssim$ 15% (partly cancels)
Association half-width	$\lesssim$ 15% (partly cancels)
2-D → arclength ordering	$\lesssim$ 15% (partly cancels)
Distance revision	−3.5%
Net (all toggled)	×1.48 (0.212→0.313 pc)

Appendix F); the thermal (f) and magnetic (mass-to-flux) criticalities are physically distinct.

We adopt a single reference $W_{\rm fil} = 0.10$ pc for comparability with prior HGBS work, though its universality is debated (Panopoulou et al. 2017, 2022); the directly-measured region-Plummer widths (0.11–0.16 pc, $\lambda/W \approx 1.4$; Section 2.4) show the sub-classical conclusion is robust to the convention, carried as the $\sim 25\%$ width systematic (Section 2.11).

3.2 Width normalisation: $W_{\rm core}$ to $W_{\rm fil}$

Three widths enter the simulation–observation comparison and must be kept separate (Fig. 10; Table 13): the initial Gaussian half-width $W_{\rm core} = 0.3\,\lambda_{\rm J}$; the formation FWHM $W_{\rm form} = 0.683\,\lambda_{\rm J}$ of the evolved projected profile ($\simeq 2.28\,W_{\rm core}$, slightly below the initial $2.355\,W_{\rm core}$ owing to contraction; resolution- and seed-converged to $< 1\%$); and the observed HGBS width $W_{\rm fil}$, the beam-convolved Plummer FWHM (Arzoumanian et al. 2019), which we obtain by forward-modelling the simulated filaments through the

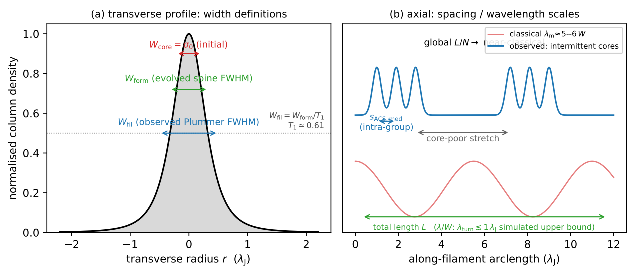

Figure 10. Schematic of the length-scale definitions used throughout (Table 1). *(a)* Transverse column-density profile, marking the initial half-width $W_{\rm core} = \sigma_0$, the evolved spine FWHM $W_{\rm form}$, and the observed beam-convolved Plummer FWHM $W_{\rm fil} = W_{\rm form}/T_1$ ($T_1 \simeq 0.61$; Eq. 6). *(b)* Axial view contrasting the classical widely-spaced comb ($\lambda_{\rm m} \approx$ 5–6 W) with the observed intermittent cores: a short intra-group adjacent spacing $s_{\rm ACS,med}$ within core-rich stretches separated by core-poor filament, whose filament-averaged line density L/N is near-classical, while the simulated fragmentation scale is an upper bound $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$.

full synthetic-HGBS pipeline (projection, beam convolution, Plummer fit). This gives $\eta_W = W_{\rm form}/W_{\rm fil} = 0.59$–$0.78$ and

$$T_1 \equiv \frac{W_{\rm form}}{W_{\rm fil}} = 0.61 \quad \text{(forward-model band 0.59–0.78)}, \quad (5)$$

so that $W_{\rm fil} = W_{\rm form}/T_1 \approx 1.12\,\lambda_{\rm J}$. We quote the forward-model best-fit $T_1 = 0.61$; because it lies near the lower edge of the band, a mid-band $T_1 \approx 0.68$ would give a slightly *larger* (less sub-classical) $\lambda/W_{\rm fil}$, so this normalisation is deliberately not the headline; the primary comparison uses the width-convention-independent growth-rate upper bound (Section 5), and this T_1 conversion is retained only as a cross-check. The observable is $\lambda/W_{\rm fil}$; because the fragmentation spacing λ is a physical length independent of the width

Table 13. The three filament widths used in the simulation–observation comparison (all in Jeans lengths, $\lambda_{\rm J} = 1$).

Symbol	Value ($\lambda_{\rm J}$)	Definition
$W_{\rm core}$	0.30	initial Gaussian half-width σ (input)
$W_{\rm form}$	0.683	Gaussian FWHM of evolved projected profile, supercritical ensemble ($\simeq 2.28\, W_{\rm core}$)
$W_{\rm FWHM}$	0.58	Gaussian FWHM at the growth-rate epoch, near-critical runs (narrower, already contracting); normalises Table C1
$W_{\rm fil}$	≈ 1.12	beam-convolved Plummer FWHM ($= W_{\rm form}/T_1$); the observable normalisation

definition, the conversion from the simulation-internal ratio $\lambda/W_{\rm core}$ (normalised by the initial $\sigma = 0.3\,\lambda_{\rm J}$) is

$$\left(\frac{\lambda}{W}\right)_{\rm fil} = \frac{W_{\rm core}}{W_{\rm fil}}\left(\frac{\lambda}{W}\right)_{\rm core} \approx 0.27\left(\frac{\lambda}{W}\right)_{\rm core}, \tag{6}$$

i.e. $\lambda/W_{\rm fil} = \lambda\,[\lambda_{\rm J}]/1.12$; the prefactor is $0.27 = W_{\rm core}/W_{\rm fil} = 0.30/1.12$ (equivalently T_1 times the $\sigma \to$FWHM factor, $0.61 \times 0.44 = 0.27$, using $T_1\, W_{\rm core}/W_{\rm form}$), and using T_1 alone would overstate $\lambda/W_{\rm fil}$ by $\approx 2.3\times$. This prefactor is *not* a fixed constant: it carries the full $\approx 30\%$ T_1 band ($T_1 = 0.59$–0.78), so 0.27 is the best-fit value of a ≈ 0.26–0.34 range. Because the observable is sensitive to how $W_{\rm fil}$ is defined, the primary comparison uses the width-convention-independent growth-rate wavelength (an upper limit; Section 5), and this normalisation is retained only as a cross-check.

For the ambient-confined supercritical ensemble ($\lambda/W_{\rm core} = 3.27$) this gives $\lambda/W_{\rm fil} \approx 0.9$.

3.3 Magnetic Tension Mechanism

For a filament threaded by a magnetic field at an angle θ to the axis (Alfvén speed v_A), a schematic oblique dispersion relation, valid only for a static background field, is (Nakamura et al. 1993):

$$\omega^2 = c_s^2 k^2 + v_A^2 k^2 \sin^2\theta - 4\pi G \rho_0 F(kR) \tag{7}$$

where $F(kR)$ encodes the cylindrical geometry and θ is the field–axis angle. This is a *schematic static-field* relation that treats $\mathbf{B}$ as a fixed background and *cannot be used quantitatively* for a magnetised cylindrical equilibrium; in particular it does not describe a current-carrying (helical) equilibrium. We use it only as an analytic guide, so this paper contains *no quantitative model of magnetic tension*; it nonetheless fixes the *sign* of each idealised case, the magnetic term $v_A^2 k^2 \sin^2\theta$ vanishing for a *longitudinal* field (reducing to the field-free thermal relation) and being maximal for a *perpendicular* one ($\theta = 90^\circ$, which lengthens the axial mode). Consistent with this, our longitudinal runs keep the growth-rate wavelength at order $\lambda_{\rm J}$ across $\beta = 0.3$–1.0 ($f = 1.1$; the $\beta = 1.0$ ladder gives $\lambda_{\rm turn} = 1.14\,\lambda_{\rm J}$, Table C1) with no resolved β-trend, so the scale is set by the thermal Jeans length, not the field, and the shortening we measure is the dynamical/contraction shift, not magnetic tension. A genuinely shorter magnetic wavelength would require a force-balanced *helical*/toroidal field, whose stability is a cylindrical magnetostatic-equilibrium problem that Eq. (7) cannot address. The full case-by-case discussion of the longitudinal, perpendicular and helical geometries, and the sense in which the simulations *constrain only the specific initial field geometries simulated* rather than dispose of magnetic alternatives, is deferred to Section 6.3. The seeded helical run tested here is an *out-of-equilibrium pinch*, not a force-balanced equilibrium, and does *not* rule out the helical mechanism (Section 4.3).

3.4 Hierarchical Fragmentation

Filament fragmentation is hierarchical: filaments fragment into velocity-coherent fibres, and fibres into cores (Hacar et al. 2013, 2018; Yang et al. 2024). Whether the fibre-to-core spacing is quasi-periodic at the classical scale is not established; the ALMA study of Yang et al. (2024) finds a clear filament–fibre–core hierarchy but only a weak quasi-periodic imprint at the core level. If HGBS filaments are bundles of fibres each fragmenting near the classical scale, measuring core spacings across the whole filament compresses the apparent spacing, in the observed direction. We test this quantitatively in Section 6.2, where a *toy geometrical model* of N-fibre bundles run through our ACS pipeline reproduces the observed λ/W; it emerges as a viable but not unique explanation (single-filament dynamical fragmentation is also consistent), and confirming it requires fibre-resolved core-spacing analysis beyond the single Orion B region of Yang et al. (2024).

4 MHD SIMULATIONS

The full campaign comprises 2251 Athena++ runs across 18 grids (Table E1), consuming of order 10^5 CPU-hours on Google Cloud E2 infrastructure; controlled numerical experiments, not models of individual HGBS filaments. The large majority are *exploratory* runs that map outcomes and timescales; the headline *wavelength* result rests instead on the load-bearing *production* runs (the 25-run near-critical growth-rate campaign, Table C1, and its 8-run 128^3–512^3 convergence ladder; Table 14), so the 2251-run total is *not* the statistical weight behind the wavelength. Two results emerge (Section 6.3): the *outcome* of a run (radial collapse or longitudinal beading) depends on regime and boundary conditions, so a single filament has no setup-independent fate, whereas its dominant fragmentation *mode* is a sub-classical upper bound (≈ 0.5–$1\times$ the classical value; Section 5) for near-critical, longitudinal-field filaments (the perpendicular geometry is not numerically resolved). This section provides the evidence for both.

4.1 Simulation Methodology

We performed self-gravitating simulations with Athena++ (Stone et al. 2020), solving the standard ideal-MHD equations, continuity, momentum with the Lorentz force, ideal induction, and total energy, coupled to self-gravity through the Poisson equation $\nabla^2\phi = 4\pi G\rho$.

We initialise a cylindrical filament with density profile $\rho(r) = \rho_0/[1 + (r/W)^2]$ and uniform field $\mathbf{B}_0$, characterised by the thermal line-mass fraction $f = M_{\rm line}/M_{\rm line,crit}$ (Eq. 2), plasma $\beta = 8\pi P/B^2$, and the seed-amplitude parameter $M_{\rm seed} = \delta v/c_s \times 10^4$. At filament conditions ($n \sim 10^4\,{\rm cm^{-3}}$,

Table 14. Load-bearing *production* simulations underlying the fragmentation-wavelength result: the near-critical growth-rate ladder (campaign 15) and the direct supercritical ensemble (campaign 11). Shared parameters (f, $\mathcal{M}$, β, θ, boundary conditions and collapse/bead time) are given in each block sub-header; per-row entries reuse the growth-rate measurements of Table C1. These production runs, not the 2251-run exploratory total, carry the quoted $\lambda_{\rm turn}$.

f	seed	resolution	$\lambda_{\rm turn}/\lambda_{\rm J}$	$\lambda_{\rm turn}/W$
Near-critical growth-rate ladder (campaign 15): β=1.0, θ=0°, $\mathcal{M}$=10^{-3} broadband seed, ambient BC (periodic/reflecting), bead $t_{\rm frag}$≈1.1–1.2 $t_{\rm J}$				
1.1	42	128^3	1.14	1.95
1.1	42	256^3	1.14	1.99
1.1	42	512^3	1.14	2.00
1.2	42	256^3	1.00	1.81
1.2	137	256^3	0.89	1.60
1.5	42	256^3	0.89	1.47
1.5	137	256^3	0.89	1.47
1.5	42	512^3	0.89	1.47
1.8	42	256^3	0.80	1.19
2.0	42	256^3	0.80	1.20
Direct supercritical ensemble (campaign 11, 39 runs): f=1.5–2.0, β=0.5–2.0, θ=0°, seeds 42/137, periodic+reflecting ambient BC, bead $t_{\rm bead}$≈0.6–0.75 $t_{\rm J}$				
ensemble		256^3	1.0–1.14	0.89–1.01†

† In beam-convolved Plummer units, $\lambda/W_{\rm fil}$ (KS $p = 0.17$ between periodic and reflecting BCs); the ladder's $\lambda_{\rm turn}/W$ column is in the Gaussian formation FWHM of Table C1.

$T \sim 10\,$K) $\beta = 1$–0.3 corresponds to $B \sim 15$–$30\,\mu$G, the range of dense-filament estimates (Crutcher 2012). $M_{\rm seed}$ is a perturbation-spectrum label, not a physical Mach number: the base grid uses a single-mode seed $\delta v/c_s = 10^{-4}$, whereas the growth-rate ladder of Section 5 uses a 10^{-3} twelve-mode broadband seed. The magnetic mass-to-flux ratio scales as $\mu_\Phi/\mu_{\Phi,{\rm crit}} \propto f\sqrt{\beta}$ and exceeds unity for every configuration (none strictly magnetically subcritical), so the low-β runs are magnetically *supported*.

We explored a base grid ($\mathcal{M} = 0.5$–5, $\beta = 0.05$–5, near-critical) and a transition grid ($f = 1.4$–2.0, $\beta = 0.3$–1.3, $\mathcal{M} = 1$–5) to resolve the stable–unstable boundary; full ranges, run counts and the resulting three outcome regimes (magnetically supported quasi-stability at $\beta \lesssim 0.15$, radial collapse, and longitudinal fragmentation) are deferred to Appendices E and F. The near-critical longitudinal case that is our focus beads at $t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$, turbulence changing the *wavelength* by $< 5\%$ while shortening the timescale 3–5×. The load-bearing *production* runs on which the quoted wavelength rests, the near-critical growth-rate ladder and the direct supercritical ensemble, are collected in Table 14. These runs satisfy the Truelove criterion at the spacing-setting epoch ($N_J \approx 5$–7 cells per local Jeans length, in both the axial and transverse directions on the isotropic cubic grids) but sit close to the numerical floor, only 1.25–1.75× the $N_J = 4$ minimum, which is one reason we quote only an order-$\lambda_{\rm J}$ upper bound (Appendix D).

4.2 Ambient confinement and the direct supercritical measurement

The outcome for a supercritical filament is set by whether it is confined by an ambient medium, as real HGBS filaments are (d denotes the axis-to-boundary distance in Jeans lengths). The main 654-run grid has no ambient medium, so its Gaussian skirt feeds an accretion channel through the periodic boundary that suppresses beading and drives radial collapse (Appendix F1); with ambient confinement (`filament_ambient` generator) the same configurations fragment longitudinally, a 12-run reconciliation suite beading at physical, β-dependent collapse times (0.55–0.70 $t_{\rm J}$) that converge across boundary conditions.

Ambient-confined supercritical filaments ($f = 1.5$–2.0, $\theta = 0°$) fragment longitudinally: an ensemble of 39 runs (periodic and reflecting BCs, ambient confinement) develops 3–10 interior axial density maxima at a spacing of order the Jeans length. We take the wavelength from the linear growth-rate spectrum, whose low-n modes are resolution-stable: no physical turnover is detected above $\approx 1\,\lambda_{\rm J}$, so the fastest-growing wavelength is constrained only to $\lambda_{\rm turn} \lesssim 1.0$–$1.14\,\lambda_{\rm J}$ (an upper bound; no resolved peak), $\lesssim 0.35$–0.5 times the classical value (not the median peak count, which is epoch- and resolution-dependent). The ambient medium confines the filament (central-to-ambient density ratio $\sim 10^2$, consistent with HGBS crest-to-background contrasts; Arzoumanian et al. 2019), so it is representative rather than tuned. Across the 39 runs periodic and reflecting boundaries are statistically indistinguishable (KS $p = 0.17$; medians $\lambda/W_{\rm fil} \approx 0.89$ and 1.01), and this ensemble median and the growth-rate result are the same physical wavelength $\lambda \approx 1.1\,\lambda_{\rm J}$ in different normalisations. The ensemble beads early ($t_{\rm bead} \approx 0.6$–$0.75\,t_{\rm J}$), before runaway and at moderate contrast, so with $\lambda \approx \lambda_{\rm J}$ resolved by ∼32 cells it satisfies Truelove at measurement (Appendix D).

Perpendicular-field filaments ($\theta = 90°$) do not universally spindle in the raw low-resolution grids: an 18-run ideal-MHD grid at $d = 1.0$ shows 7/18 bead (at $\beta = 0.5$ and 2.0) with an apparently short spacing ($\lambda/W_{\rm fil} \approx 0.4$), but this conflicts with static linear theory (which predicts a *longer* mode) and a transverse-refinement ladder to $N_J \approx 170$ resolves the near-critical case into a monolithic radial spindle, identifying the short beading as a Truelove artefact (Section 4.3). An exploratory ambipolar-diffusion survey was also run, but it fragments only at $N_J \approx 1$–2, below the Truelove limit, so it cannot support any conclusion and we exclude it from the argument here; it is described, with an explicit warning that it is numerically unresolved, in Appendix F3.

4.3 Testing the resolution and magnetic-field alternatives

Three targeted campaigns test the concerns that most affect the interpretation: whether the longitudinal wavelength is resolution-limited, whether the perpendicular geometry produces genuine short beading, and whether a helical field, the one configuration expected to shorten the axial mode, actually does.

Resolution stability of $\lambda_{\rm turn}$. Two refinement directions behave oppositely. Under *transverse* refinement (the isotropic 128^3–512^3 ladder) the short-wavelength spectrum ($\lambda \lesssim$

$0.9\,\lambda_{\rm J}$, $n \gtrsim 9$) grows with resolution and is numerical (unresolved transverse sub-fragmentation; Fig. 14a), whereas the physical band ($\lambda \gtrsim 1\,\lambda_{\rm J}$, $n \leq 8$) is converged. Under *axial-only* refinement ($512 \times 128^2 \to 1024 \times 128^2$) the physical band reproduces to better than 0.5% (Γ at $n = 6, 7, 8$, i.e. $\lambda = 1.33, 1.14, 1.00\,\lambda_{\rm J}$, is 17.6, 18.7, 18.7 at both resolutions, plateauing without turning over), *and* the short-λ rise is unchanged, confirming it is fixed by the axial Nyquist scale, not a physical mode migrating to shorter λ. The $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$ upper bound is therefore stable: axial refinement does not move the physical-band plateau, and the only shorter-λ structure is the transverse-resolution artefact already identified.

The perpendicular geometry: a resolved spindle, not short beading. On a transverse-refinement ladder (128^3–512^3) the near-critical *ideal* perpendicular filament collapses monolithically in the radial direction at every resolution, the axial mode staying at the box scale and the axial perturbation at seed level throughout the well-resolved phase (at 512^3 the spine stays resolved with $N_J \approx 170$ cells at $C \approx 3$, falling only to $N_J \approx 40$ at $C \approx 30$; still $\approx 10\times$ above Truelove; Appendix D). Short-wavelength axial structure ($\lambda/W \approx 0.4$) appears only once the spine drops below $N_J \approx 1$; refinement delays its onset to deeper collapse (Fig. 11), the signature of a grid artefact. The perpendicular field therefore *suppresses* axial fragmentation in favour of radial collapse, as static linear theory predicts (a perpendicular field lengthens the axial mode), and the short low-resolution beading is numerical. This is for *ideal* MHD; the ambipolar-perpendicular case (Section 4.2) remains separately unconverged.

A seeded helical field pinches the width but does not shorten $\lambda_{\rm turn}$*; the force-balanced case remains untested.* A divergence-free helical-field scan ($B_\phi/B_z = 0$–2; construction and scan detail in Appendix F3) leaves the axial growth-rate plateau at $\lambda \approx 1\,\lambda_{\rm J}$ for every winding and produces only a radial *pinch* that narrows $W_{\rm FWHM}$ by $\approx 25\%$ (raising λ/W by reducing W, not by shortening λ; Fig. 12), so, being seeded rather than force-balanced, it cannot test the confining Fiege–Pudritz sausage ($m = 0$) mode (Fiege & Pudritz 2000); the one geometry that could shorten λ, which therefore remains the single relevant magnetic configuration left unconstrained (Section 6.3).

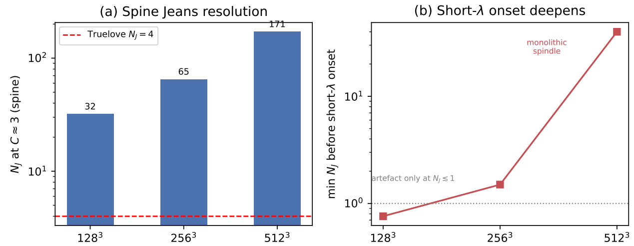

Figure 11. Ideal perpendicular-field refinement ladder (128^3–512^3; Section 4.3). *(a)* Cells per local Jeans length N_J at the beading-relevant spine contrast $C \approx 3$: the collapsing spine is resolved far above the Truelove threshold ($N_J = 4$, dashed) at every resolution. *(b)* The spurious short-λ axial "beading" emerges only once the spine drops below $N_J \approx 1$, and refinement pushes that onset to ever-deeper collapse (at 512^3 the filament stays monolithic to $N_J \approx 40$ at $C \approx 30$); the short-λ onset deepening with resolution being the textbook Truelove under-resolution artefact signature, not a physical fragmentation mode.

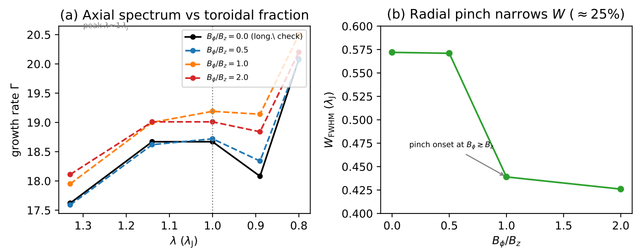

Figure 12. Divergence-free helical-field scan ($B_\phi/B_z = 0$–2, 512×128^2; Section 4.3). *(a)* The axial growth-rate spectrum $\Gamma(\lambda)$ in the physical band is invariant to the toroidal fraction ($< 6\%$, fit noise), with the plateau peak fixed at $\lambda \approx 1\,\lambda_{\rm J}$; $B_\phi/B_z = 0$ (black) reproduces the pure-longitudinal spectrum, a code check. No excess growth appears at shorter λ for any winding. *(b)* The one systematic effect is a radial *pinch*: the formation FWHM narrows by $\approx 25\%$, with onset at $B_\phi \gtrsim B_z$. The seeded helical field therefore raises λ/W by shrinking W, not by shortening λ; the force-balanced Fiege & Pudritz (2000) pinch mode is not tested here.

5 THE FRAGMENTATION SCALE AND ITS COMPARISON WITH OBSERVATION

The measured core spacing is sub-classical only when expressed against the filament FWHM, and that comparison hinges on how the width is defined. We therefore anchor it on the spacing in units of the filament FWHM measured the same way on simulations and data, and verify the whole chain end to end (Appendix C).

First, the measurement recovers injected spacings: passing synthetic filaments with a known spacing and width through the identical pipeline returns λ/W to better than 0.5% over $\lambda/W = 1.4$–5.7 (Appendix C), so the observed→simulated comparison depends on the width convention, not on residual measurement bias. Second, forward-modelling the simulations end to end through the HGBS pipeline (projection, beam convolution, Plummer fitting, and the same ACS measurement) gives the observable $\lambda/W_{\rm fil}$ directly. Two configurations bracket the result: the *near-critical* growth-rate ladder forward-models to $\lambda/W_{\rm fil} \approx 1.0$–1.3, while the *supercritical* 39-run bead ensemble gives ≈ 0.89–1.01 (Table 14); both are order-$\lambda_{\rm J}$ and both exceed the analytic shortcut (≈ 0.9, Eq. 6, which applies only a single $\sigma \to$FWHM factor). We adopt the near-critical forward-model band (1.0–1.3) as the headline like-for-like value and carry the $\approx 30\%$ spread between it and the analytic/ensemble estimates as the width-conversion systematic.

The fragmentation wavelength must be measured with care: counting density peaks late in the collapse is epoch- and resolution-dependent, so we instead identify the dominant mode from the growth-rate spectrum. Seeding the filament with a small broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve axial modes), we track the power $P(k,t)$ in each mode and fit its growth over its approximately exponential interval (Fig. 13); the growth accelerates as the filament collapses, so we isolate the dominant mode of a *dynamically collapsing* filament. The growth rate $\Gamma(\lambda)$ (Fig. 14, left) rises as λ decreases

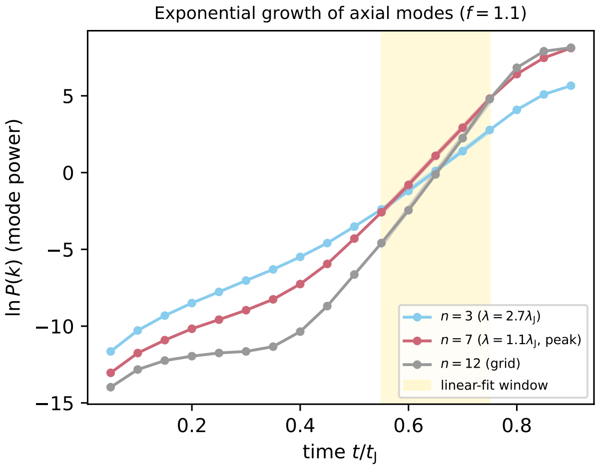

Figure 13. Power in three axial modes versus time for a near-critical filament ($f = 1.1$). Each grows with an approximately exponential interval (the fit window, shaded) whose slope gives the growth rate; the growth steepens as the filament collapses, so the extracted rate characterises the dominant mode of a dynamically collapsing filament. The peak mode $n = 7$ grows faster than the long-wavelength $n = 3$, giving the sub-classical $\lambda_{\rm turn}$; the grid-scale $n = 12$ is shown for reference.

and is resolution-stable for $\lambda \gtrsim 1\,\lambda_{\rm J}$ (low-n modes agree to $\sim 5\%$ across 128^3–512^3; Γ at $n = 7$ is $17.8, 18.5, 18.7$), rising to a *plateau* at the converged-band edge (Γ at $n = 6, 7, 8$, i.e. $\lambda = 1.33, 1.14, 1.00\,\lambda_{\rm J}$, is $17.6, 18.7, 18.7$), while below $\lambda \lesssim 0.9\,\lambda_{\rm J}$ it rises with *transverse* resolution and is numerical. Because Γ plateaus rather than turning over, we quote the dominant wavelength as an *upper bound*, $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$, and do *not* claim a precisely-located maximum (Appendix C); the true peak could lie at shorter, transverse-unresolved wavelength ($\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$), so we use $\lambda_{\rm turn}$ below only as shorthand for this upper bound. It is an order-$\lambda_{\rm J}$, sub-classical value: $\lambda_{\rm turn}/W_{\rm FWHM} \approx 1.2$–$2.0$ (near-critical 1.5–2.0, falling to ≈ 1.2 in the supercritical runs; Table C1), a factor ≈ 2.5–5.0 below the classical $\lambda_{\rm m} \approx 5$–$6\,W_{\rm FWHM}$ (in W units; Fischera & Martin 2012) and at or below the critical $2.6\,W_{\rm FWHM}$ across $f = 1.1$–2.0. The like-for-like beam-convolved Plummer value (below) is $\lambda/W \approx 1.0$–1.3, and it is this that should be set against the data (Table C1).[2]

The result is explicitly one-sided: the simulations rule out the assertion that a contracting filament *must* fragment only at the equilibrium-cylinder wavelength, but they do *not* determine the physical minimum or preferred wavelength, and we do not claim the numerical value predicts the observed spacing; only that the two are compatible. The convention-independent statement is that $\lambda_{\rm turn}$ of a dynamically collapsing filament is sub-classical: in width-free form $\lambda_{\rm turn} \lesssim 1.0$–$1.14\,\lambda_{\rm J}$ versus the classical $\lambda_{\rm m} = 22H = 2.3\,\lambda_{\rm J}$ (a factor ≈ 2), which we take as primary. This is an *upper bound* ($\lambda_{\rm turn}/\lambda_{\rm m} \approx 0.35$–$0.5$, wavelength ratio, across $f = 1.1$–2.0): the spectrum plateaus without turning over, and the selected *mode* is imprinted early and fit-window-invariant (Appendix C), by which epoch $C \approx 20$–40 and $N_J \gtrsim 5$. The load-bearing near-critical measurement has $N_J \approx 5$–7 cells per local Jeans length at the spacing-setting epoch, i.e. 1.25–$1.75\times$ the Truelove ($N_J = 4$) minimum, and the growth-rate fit window ($\delta_{\rm rms} < 0.05$) keeps $N_J > 4$ at all three resolutions (Appendix D). The Truelove criterion guards against *artificial* fragmentation; it is a necessary minimum, not a guarantee of a few-per-cent-accurate dispersion relation. We therefore quote only an order-$\lambda_{\rm J}$ upper bound and do not claim $1\,\lambda_{\rm J}$ to a few per cent. To be explicit about what the simulations do and do not establish: they demonstrate an order-$\lambda_{\rm J}$ *upper bound* and the *absence* of the required classical-scale (5–$6\,W$) maximum, *not* a converged fastest-growing wavelength; $1.0\,\lambda_{\rm J}$ is a band edge, not a converged physical maximum. Nor is the no-turnover result a coarse-k (mode-quantisation) artefact: the three-resolution ladder (128^3–512^3) resolves a different maximum axial wavenumber at each resolution, yet the converged-band plateau and its edge do not move, decoupling the axial mode quantisation from the resolution limit.

Two caveats temper the causal attribution. First, the fitted rates ($\Gamma \approx 18$–$26\,t_{\rm J}^{-1}$) are the accelerating growth of a collapsing background, not quasi-static eigenvalues, so we use only the mode *ordering*. Second, a $p = 4$ (isothermal-equilibrium) profile gives the same order-$\lambda_{\rm J}$ wavelength ($\approx 0.8\,\lambda_{\rm J}$; Appendix C), so the deficit is not a profile-index artefact but a generic outcome of fragmentation from a contracting state (Section 6.4).

The clean physical reading follows from evaluating the classical scale at the *instantaneous central density*, $22H(\rho_{\rm eff}) \approx 0.8\,\lambda_{\rm J}(\rho_0)$: the instability remains approximately classical with respect to the instantaneous central density. The sub-classical ratio λ/W arises because the observable FWHM does not contract in step with ρ_c, since the extended envelope lags the collapsing spine. Consistent with this, $\lambda_{\rm turn} \lesssim 1.1\,\lambda_{\rm J}$ sits at or below the analytic *cutoff* $\lambda_{\rm cr} \approx 1.2\,\lambda_{\rm J}$ of our own eigensolver (Appendix C), not the classical *peak* $2.3\,\lambda_{\rm J}$: this is the expected signature of a contracting filament sliding towards marginal stability, not a re-measurement of the equilibrium-cylinder cutoff (the growing modes are near the marginal scale of the *contracting*, elevated-density state). The equilibrium-recovery control confirms this reading, returning the classical 5–6 when the filament instead begins from equilibrium.

A time-dependent mode-selection estimate. This picture can be sharpened from a counterexample into an approximate *consistency check* (not an independent prediction, since the freeze-out density is read from the measured growth rates) by a WKB/self-similar treatment of fragmentation during contraction, in the spirit of Larson (1985) and Clarke et al. (2016, 2017). As the filament contracts, its central density $\rho_c(t)$ rises, so the instantaneous classical fastest-growing wavelength shrinks as $\lambda_{\rm m}(t) = 22H[\rho_c(t)] = 2.3\,\lambda_{\rm J}(\rho_0)\,(\rho_c/\rho_0)^{-1/2}$, while the instantaneous growth rate rises as $\Gamma \propto (4\pi G\rho_c)^{1/2}$. A perturbation is *selected*, imprinted non-linearly, not at the marginally-stable equilibrium scale but at the epoch $t_\star$ when the fastest-growing mode completes its $\sim N_e$ e-foldings from seed to non-linearity within the residual contraction time (the standard collapse-fragmentation condition $\tau_{\rm grow} \lesssim \tau_{\rm contract}$; Larson 1985). This fixes a *freeze-out* density $\rho_{\rm eff}$, which we

[2] The near-critical formation FWHM is $W_{\rm FWHM} = 0.58\,\lambda_{\rm J}$, narrower than the supercritical $W_{\rm form} = 0.683\,\lambda_{\rm J}$ because a near-critical filament has already begun to contract when its mode is set (both tabulated in Table 13).

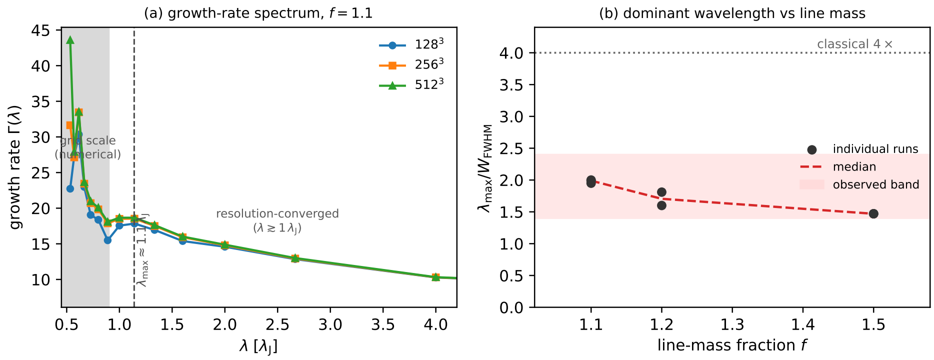

Figure 14. *(a)* Growth-rate spectrum $\Gamma(\lambda)$ of the axial modes for a near-critical filament ($f = 1.1$) at three resolutions. For $\lambda \gtrsim 1\,\lambda_{\rm J}$ the curves overlap (resolution-converged); at the grid scale ($\lambda \lesssim 0.9\,\lambda_{\rm J}$, shaded) they separate and rise with resolution, the signature of unresolved sub-fragmentation. Γ plateaus (does not turn over) at the converged-band edge, so $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$ is an *upper bound*, not a resolved maximum. *(b)* $\lambda_{\rm turn}/W_{\rm FWHM}$ versus line-mass fraction (Gaussian FWHM), well below the classical 5–6 at every f. The like-for-like comparison uses a beam-convolved Plummer width $\sim 1.5\times$ broader, giving matched $\lambda/W \approx 1.0$–1.3 (grey band): this, not the Gaussian ≈ 2, is what should be set against the observed $\lambda/W \approx 0.4$–1.9.

infer from the measured rates: $\Gamma_{\rm meas} \approx 18$–$26\,t_{\rm J}^{-1}$ versus the base-density value $\Gamma(\rho_0) \approx 6.3\,t_{\rm J}^{-1}$ (Appendix C) gives $\rho_{\rm eff}/\rho_0 = (\Gamma_{\rm meas}/\Gamma(\rho_0))^2 \approx 8$–$17$; this is corroborated *independently* by the directly-measured central-density contrast at the mode-setting epoch ($\rho_c/\rho_0 \approx 8$–17; Appendix C), which de-circularises the estimate. The instantaneous classical relation then gives $\lambda_{\rm select} = 2.3\,\lambda_{\rm J}(\rho_0)/\sqrt{\rho_{\rm eff}/\rho_0} \approx 0.6$–$0.8\,\lambda_{\rm J}$, which *abuts* (and slightly under-predicts) the measured upper bound $\lambda_{\rm turn} \approx 0.8$–$1.1\,\lambda_{\rm J}$, consistent in sign, and in size only where the ranges overlap ($\approx 0.8\,\lambda_{\rm J}$). The same argument predicts $\lambda_{\rm select}$ to *decrease* with contraction rate, more supercritical filaments contract faster, reach higher $\rho_{\rm eff}$, and freeze at shorter λ; consistent with the f-ladder ($f = 1.1$: $\Gamma \approx 18$, $\lambda_{\rm turn} = 1.14$; $f = 1.8$–2.0: $\Gamma \approx 25$, $\lambda_{\rm turn} = 0.80\,\lambda_{\rm J}$; Table C1), although the $f = 1.8$–2.0 values lie below the converged band edge and are themselves unconverged upper bounds, so this is an upper-bound trend rather than a measured dispersion relation. The sub-classical *ratio* follows because the spine density rises by $\rho_{\rm eff}/\rho_0 \approx 8$–$17$ (so H falls by $\sqrt{8\text{–}17} \approx 3$–$4$) while the observable envelope FWHM contracts only modestly ($W_{\rm form} \approx 0.58$–$0.68\,\lambda_{\rm J}$, a factor $\lesssim 1.3$): the deficit is precisely this spine–envelope contraction lag. This is a falsifiable scaling, $\lambda_{\rm select} \propto \rho_{\rm eff}^{-1/2}$ with $\rho_{\rm eff}$ set by the contraction rate, that a future relaxation/contraction ladder can test directly.

Single-filament gravitational fragmentation, evolved dynamically, thus produces a converged *upper bound* that is sub-classical and sits within the methodological envelope of the observed spacing (Fig. 9; within the factor-$\sim$2 width and skeleton systematics) for the near-critical filaments dominating the sample; the robust deliverable is the density-independent ratio λ/W with its equilibrium-recovery calibration. That control is the load-bearing evidence (Fig. 15): the *identical* pipeline recovers the classical turnover at $22H$ for a drag-relaxed equilibrium cylinder ($\lambda_{\rm max}/22H = 1.07$–$1.10$, both resolutions; a genuinely resolved peak) but finds no turnover for the contracting runs. It is a two-point comparison (one relaxed equilibrium versus one contracting run), so it establishes the *sign* of the effect, not its functional dependence on the contraction rate; a relaxation ladder to map that dependence is deferred (Section 6.4). This is a numerical instance of the generic expectation that a still-contracting filament fragments below the marginally-stable hydrostatic mode (Clarke et al. 2016, 2017, 2020), not a new mechanism; our distinct contributions are the longitudinal near-critical demonstration, that control, and the homogeneous Gaia-DR3 reanalysis. We note the limits of this correspondence: the single-filament simulations produce a *quasi-periodic* axial comb (3–10 near-uniformly spaced beads), whereas the observed cores are non-periodic and spatially intermittent (Section 2.10), rejecting both periodic and Poisson models. The simulations therefore reproduce a *scale* (sub-classical, order $\lambda_{\rm J}$) but neither the observed point process nor its non-periodicity; they establish only that non-equilibrium gravitational fragmentation *can* generate structure below the equilibrium-cylinder wavelength, not that it produces the observed spatial statistics.

The physically appropriate comparison is between the simulated fragmentation scale, which produces *one bead per wavelength*, and the observed *direct line density* $(L/N)/W \approx 3.4$–5.8 (Section 2.8), the quantity a one-fragment-per-wavelength theory actually predicts, rather than the intra-group $s_{\rm ACS,med}/W$. On L/N the simulated upper bound ($\lambda/W_{\rm fil} \approx 1.0$–$1.3$) is in fact *more* discrepant: the simulation makes about one bead per $\lambda_{\rm turn}$, i.e. a denser core production than the observed filament-averaged L/N, so single-filament fragmentation *over*-produces cores per unit length unless most of the filament is core-poor; precisely the spatial-intermittency picture (Section 2.10). The shorter intra-group $s_{\rm ACS,med}/W \approx 1.4$ is then the *intra-group clustering scale*, not the global fragment-production comparison, so the near-coincidence of the simulated 1.0–1.3 with it is not the key agreement. Single-filament dynamical fragmentation is thus at best marginally sufficient: it establishes consistency, not sufficiency, and hierarchical-fibre projection (Section 6.2) and magnetic tension remain viable within the factor-$\sim$2 systematics. The claim is falsifiable: it is refuted if resolved single filaments or individual fibres show a core line-density no higher than the classical one-core-per-$\lambda_{\rm m}$, and disfavoured if the true observed λ/W fell well below ~ 1 (single-filament fragmentation would then *over*-produce cores), so it is operationally viable only between these limits.

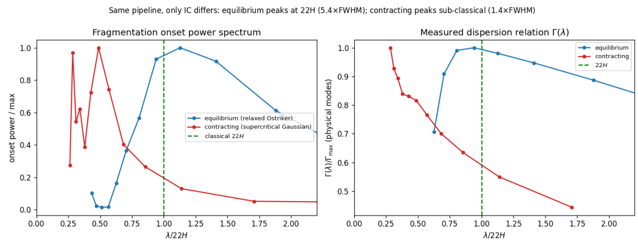

Figure 15. Equilibrium-recovery control (the primary validation that the sub-classical spectrum is not generated by the wavelength-extraction pipeline and is associated with the non-equilibrium initial state). The identical growth-rate *measurement* pipeline is applied to a *relaxed, force-balanced* pressure-confined Ostriker equilibrium (blue; prepared by explicit drag relaxation, then evolved drag-free) and to a *dynamically contracting* supercritical filament (orange); the two differ in their prepared initial state, not in the measurement, isolating the effect of starting from equilibrium versus contraction. *Left:* onset fragmentation power spectrum; the equilibrium peaks at the classical $\lambda_{\rm m} \approx 22H$ ($\approx 5.4\,W_{\rm FWHM}$) while the contracting filament peaks sub-classically ($\approx 1.4\,W_{\rm FWHM}$, upper limit). *Right:* the dispersion relation $\Gamma(\lambda)$ versus $\lambda/22H$; the equilibrium shows the classical peak-and-decline with its maximum at $22H$, whereas the contracting filament's Γ rises monotonically. The pipeline thus recovers the classical wavelength when the physics is classical.

Opacity-limited fragmentation. Because $\Gamma(\lambda)$ does not turn over within the converged band, an isothermal calculation contains no physical small-scale thermodynamic cutoff and therefore cannot predict a characteristic core mass or a unique CMF peak; a physical turnover, which would convert the isothermal plateau into a physical peak, requires a non-isothermal EOS ($\gamma_{\rm eff} > 1$ at $n \gtrsim 10^5$–$10^6\,{\rm cm^{-3}}$) or radiative physics (Section 6.1), the classical example being the opacity limit at which the gas becomes optically thick to its own cooling radiation (Rees 1976; Low & Lynden-Bell 1976). Consistent with this, our $\gamma = 5/3$ tests suppress fragmentation (Appendix D), confirming the true short-wavelength cutoff lies below our isothermal grid. Such a departure from isothermality occurs once dust–gas coupling and line cooling can no longer hold T constant against compressional heating; an effective $\gamma_{\rm eff} \gtrsim 1$ regime reached at $n \gtrsim 10^5$–$10^6\,{\rm cm^{-3}}$ for these clouds (Larson 1985; Glover & Clark 2012), above the $n \sim 10^4\,{\rm cm^{-3}}$ of the fragmenting filaments modelled here. Radiation-hydrodynamics or a piecewise polytropic EOS with $\gamma > 1$ at high density is therefore needed to set a physical fragmentation scale and a CMF peak; our isothermal suite deliberately does not include it.

6 DISCUSSION

6.1 Why the spacing matters: fragment masses and the CMF

The spacing matters because, with the line mass, it sets the characteristic fragment mass that seeds the core mass function (CMF) and ultimately the IMF: for a filament of line mass $M_{\rm line}$ fragmenting at spacing λ the mean fragment mass is $M_{\rm frag} \sim \epsilon\, M_{\rm line}\, \lambda$, and a near-critical $M_{\rm line} \approx 16\,M_\odot\,{\rm pc}^{-1}$ with $\lambda \approx 0.1$–$0.2\,{\rm pc}$ gives ~ 0.5–$1\,M_\odot$ for $\epsilon \sim 0.3$, consistent with the HGBS CMF peak while the classical 5–6×FWHM would overshoot it. This is only an order-of-magnitude *consistency check*: it is degenerate in $\epsilon M_{\rm line}$ (only the product enters) and yields a *mean*, not a peaked distribution. An *isothermal* EOS lacks a small-scale thermodynamic cutoff and therefore cannot predict a characteristic core mass or a unique CMF peak; only a non-isothermal EOS ($\gamma_{\rm eff} > 1$ at $n \gtrsim 10^5$–$10^6\,{\rm cm^{-3}}$) would convert the isothermal plateau into a physical peak (Section 5).

6.2 Hierarchical-fibre projection is consistent with the sub-classical spacing

A second, purely geometric route reaches the same sub-classical value without modified physics, and rests on independently observed substructure: HGBS filaments are bundles of N velocity-coherent *fibres* (Hacar et al. 2013, 2018; Yang et al. 2024). If each fibre fragments near the classical wavelength but core spacing is measured *across* the bundle, as any filament-level pipeline does, the interleaving of cores from N fibres compresses the apparent spacing. The premise, quasi-periodic fibre-to-core fragmentation at the classical scale, is a hypothesis, not an established measurement (Yang et al. 2024 find only a weak imprint), so we present projection as viable but unconfirmed.

We tested this with a *toy geometrical model* (not a physical forward model) that reuses our exact ACS measurement. Each fibre is a cylinder of width $W_{\rm fib}$ fragmenting at its *own* classical wavelength $\lambda_{\rm fib} \approx 5$–$6\,W_{\rm fib}$ (Fischera & Martin 2012); we populate a filament with N such fibres (random phase, $\sim 30\%$ lognormal scatter), project onto the shared spine, and measure in units of the *filament* width $W_{\rm fil}$. This assumes quasi-hydrostatic fibres, which sits uneasily with our own result (a near-critical fibre would itself fragment sub-classically, doubling the required r/N), so we treat the classical-fibre $(5$–$6)r/N$ result as an upper bound and note the fibre and single-filament pictures are not independent. Two effects compress the spacing, narrower fibres ($r \equiv W_{\rm fib}/W_{\rm fil} < 1$) and several of them, giving in the interleaving limit

$$\frac{\lambda}{W_{\rm fil}} \approx \frac{(5\text{–}6)\,r}{N}, \tag{8}$$

verified across r and N in Fig. 16. Either effect alone can produce a sub-classical value: a single fibre of half the filament width ($r = 0.5$, $N = 1$) gives $\lambda/W_{\rm fil} \approx 2.5$–$3.0$, as do two full-width fibres ($r = 1$, $N = 2$).

The observation constrains only the combination r/N, not N individually (Fig. 16): the DisPerSE and FilFinder conventions map to $r/N \approx 0.07$–0.35, so with observed fibre widths $r \approx 0.3$–0.5 (Hacar et al. 2013, 2018) the sub-classical value is reproduced by a modest fibre multiplicity. We claim no specific fibre number; the robust statement is that unresolved fibre multiplicity can reduce the filament-level adjacent spacing below the classical prediction with no modified physics.

Hierarchical projection is therefore *complementary* to, not exclusive of, single-filament dynamical fragmentation (Section 5): the two are multi-scale processes that can operate together, and unresolved fibre multiplicity is observationally *degenerate* with dynamical contraction in the filament-averaged spacing. Breaking that degeneracy requires the

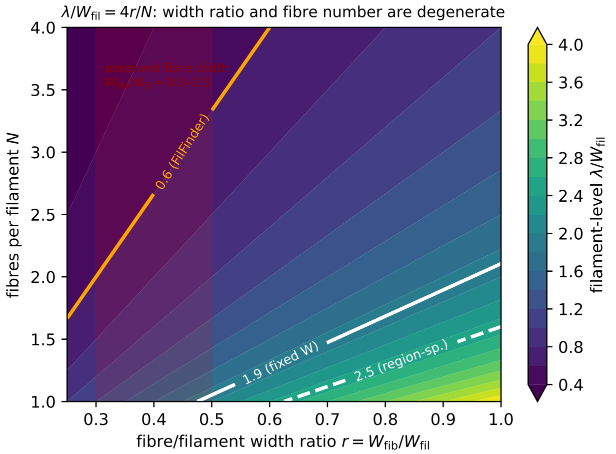

Figure 16. Hierarchical-fibre geometrical toy model, showing the filament-level $\lambda/W_{\rm fil}$ recovered by our ACS pipeline as a function of the fibre-to-filament width ratio $r = W_{\rm fib}/W_{\rm fil}$ and the number of fibres N; the model follows $\lambda/W_{\rm fil} \approx (5\text{–}6)r/N$ (Eq. 8). White contours mark the observed fixed-width and region-specific values, the orange contour the FilFinder value, and the red band the observed fibre-width range ($r \approx 0.3$–0.5). The observation constrains the combination r/N, not N alone: the sub-classical value is reproduced by a single narrow fibre, by a few wider fibres, or by intermediate combinations.

tracer that resolves the fibres directly; high-resolution kinematics (ALMA/NOEMA) that decompose the cores into their parent velocity-coherent fibres across the primary regions (Section 6.4).

A coherence point remains: our single-filament simulations fragment into a *quasi-periodic* comb at $\lambda_{\rm turn}$, whereas the observed cores are clustered with no comb (Section 2.10). The two reconcile only if sub-fragmentation, multi-fibre interleaving, and projection erase the periodicity into the observed near-exponential gap distribution, as Clarke et al. (2020) find. We do not simulate this stage, so the overlap between a single-filament $\lambda_{\rm turn}$ and the observed line density is an order-of-magnitude consistency within the methodological envelope, not a morphological match.

A third route to the same reconciliation is post-formation dynamical evolution: cores accrete, migrate along the filament and interact over several free-fall times (the migration displacements of 0.01–0.05 pc are already in our error budget; Kirk et al. 2016; Mattern et al. 2018), so dynamical relaxation of an initially quasi-periodic chain could erase the comb into the observed clustered distribution with no modified physics; a candidate on the same footing as fibre projection, testable by evolving the near-critical runs further past first fragmentation, which we have not done.

A candidate mechanism for the two-scale (local sub-classical / global near-classical) pattern. The paper's most novel empirical result, short intra-group spacing coexisting with a near-classical filament-averaged line density, calls for a physical origin, which we can only propose here. The most economical is *spatially non-uniform seeding*: real filaments inherit an inhomogeneous spectrum of density and velocity perturbations (from turbulence, accretion streamers, and fibre substructure), so gravitational fragmentation runs to non-linearity preferentially in the perturbation-rich, locally supercritical stretches, which then bead at the (sub-classical) contracting-filament scale, while low-amplitude stretches stay barren over the same interval. The result is exactly the observed picture: core-rich groups with sub-classical intra-group spacing, separated by core-poor filament, and a filament-averaged L/N close to one core per classical wavelength. Sequential/triggered fragmentation and multi-fibre interleaving (Section 6.2) produce the same two-scale signature. This is testable with the existing runs: our *uniform-seed single-filament* simulations produce a comb, not this two-scale pattern, so a direct test is to seed the near-critical runs with an inhomogeneous (e.g. turbulent) perturbation field and measure whether the simulated core chains reproduce both the sub-classical adjacent spacing *and* the near-classical L/N; a comparison we identify as the natural next step (Section 6.4).

6.3 Field geometry, boundary conditions and the perpendicular case

The common argument that most filaments carry perpendicular fields (from the Planck tendency for dense filaments to lie perpendicular to the large-scale field; Planck Collaboration et al. 2016) uses a projected orientation that need not equal the internal geometry, and hourglass morphologies with the field turning longitudinal inside the crest are observed (Jadhav et al. 2026). A resolved perpendicular filament spindles rather than fragmenting axially (Section 4.3), so it does not produce the observed axial core chains and is not a source of tension.

Where perpendicular geometry does apply, three non-exclusive resolutions of the residual deficit remain: hierarchical fibre fragmentation (bundles of near-critical fibres projected across the bundle; Hacar et al. 2013, 2018; Yang et al. 2024); non-ideal MHD at later evolution, which narrows the factor-~ 3 perpendicular deficit; and an observational lag in which the cores are relics of an earlier episode. These are separated by the future tests listed in Section 6.4.

The one untested magnetic configuration. Our field-geometry scan constrains only the specific initial field geometries simulated, the static longitudinal and perpendicular cases *within the idealised configurations tested here* (Section 4.3; Table 15), but magnetic tension is untested in its most consequential form, the force-balanced Fiege–Pudritz helical/toroidal equilibrium. The observational comparison therefore *cannot presently distinguish* dynamical contraction, hierarchical fibres or helical magnetic support, because that linear stability calculation has not been performed; its absence is not evidence against magnetic tension. An order-of-magnitude estimate read off from the published Fiege & Pudritz (2000) results makes the mechanism concrete: for their force-balanced equilibria the $m = 0$ (sausage) growth rate peaks at wavelengths of order the filament diameter when the field is strongly toroidally dominated, against $\approx 4\times$ the diameter ($\approx$ 5–6×FWHM) for the unmagnetised Ostriker cylinder; a shortening of up to a factor ≈ 2. Our HGBS-motivated range $\beta \approx 0.3$–2 (Section 3.3) puts the magnetic energy *comparable to but not dominant over* thermal, i.e. the lower end of the FP toroidal sequence, so the expected shortening is modest ($\lesssim 2$), taking the classical 5–6 W toward $\approx 3\,W$: enough to supply *part* of the observed deficit but probably not all. The precise amount for realistic pitch

Table 15. Schematic summary of the four magnetic-field configurations and their tested status.

Configuration	Tested?	Effect on axial λ
Uniform longitudinal	tested	unchanged (term vanishes)
Static perpendicular (resolved)	tested	suppresses axial fragmentation (spindle)
Seeded/dynamic helical	tested	pinches W; does not shorten λ
Force-balanced helical (Fiege–Pudritz)	not tested	could shorten by $\lesssim 2\times$ (order-of-magnitude inference from Fiege–Pudritz, not a result of this work; Section 6.3)

angles is the open calculation, requiring a validated cylindrical MHD eigensolver rather than the algebraic relation of Section 3.3; building and verifying that eigensolver is the specific obstacle, and a seeded, non-force-balanced field cannot test that mode. Concretely, a decisive future Athena++ test would (i) initialise a Fiege–Pudritz self-similar equilibrium with poloidal $B_z(r)$ and toroidal $B_\phi(r)$ satisfying radial force balance (thermal pressure + self-gravity + magnetic tension/pressure), parameterised by the toroidal-to-poloidal flux ratio and the concentration; (ii) confirm the constructed equilibrium is stationary (drag-relax, then evolve drag-free, exactly as in our equilibrium-recovery control) before any perturbation is applied; and (iii) seed the $m = 0$ (sausage) axial mode and extract $\Gamma(k)$ with the same growth-rate pipeline used here. The present seeded-helical runs test only the *unbalanced* pinch (Section 4.3), so the force-balanced hypothesis remains strictly open.

The infall connection. A dynamically fragmenting filament fragments below the equilibrium wavelength only while it is still contracting, so the interpretation ultimately rests on the filaments' accretion state. Per-cloud infall-velocity or mass-growth-rate constraints that would place Orion B, Aquila, Perseus and Taurus filaments inside or outside the simulated non-equilibrium range are not established for these specific clouds, so the simulation–cloud correspondence is a plausibility argument, not a demonstrated match.

Sub-isothermal equations of state ($\gamma < 1$) accelerate fragmentation by 30–40% without changing the wavelength, while adiabatic heating ($\gamma = 5/3$) suppresses it entirely (Appendix D). The isothermal suite is therefore the conservative case in a globally averaged sense; though not locally, where a $\sim 12\%$ dust-temperature variation shifts the local λ/W by ~ 10 per-cent and relocates where cores form (Section 6.5).

6.3.1 *Non-Ideal MHD Effects*

At the densities probed here ($n_0 \sim 10^4\,\mathrm{cm}^{-3}$), the ambipolar-diffusion time ($\sim$ 10–20 t_{ff}; Tassis & Mouschovias 2004) is long compared with the collapse time, so ideal MHD is a good approximation for the longitudinal fragmentation stage: four non-ideal runs at $\eta_{\mathrm{AD}} = \{0.01, 0.05\}$ still collapse radially, $\sim$ 12–15% *earlier* than the ideal control (flux leakage erodes magnetic support, the conservative direction). Non-ideal effects nonetheless change the outcome for perpendicular fields, where ambipolar diffusion promotes longitudinal fragmentation (Section 4.2).

6.4 Limitations and future work

Two limitations bound our conclusion. First, the characteristic λ/W and its scatter ($\sigma \approx 0.4$) rest on only four nearby, low-mass Gould Belt regions of broadly solar metallicity and similar radiation field, so the result is established only for that regime; whether it persists in high-mass complexes, at lower metallicity or under stronger irradiation is untested, and extending the pipeline to further HGBS regions and independent surveys (e.g. JCMT/SCUBA-2) is needed. Two validation checks remain *outstanding*: the $\lesssim$ 5% manual-editing bound is a *proxy* pending a blind, multi-editor re-skeletonisation, and a dedicated faint-segment injection–recovery test is not yet done. To date the adjacent-core-spacing and skeleton pipelines have *not* been independently re-run by anyone other than the author; the deposited code and catalogues are released to enable that external check. Two analyses would further sharpen the comparison: (i) a *per-segment hierarchical normalisation*; the interval-midpoint local-width test is already done (Section 2.8; medians $\Delta s/W_{\mathrm{local}} \approx$ 0.6–1.1, all sub-classical), but the cleaner unbranched-segment version is deferred because the bridged skeletons over-fragment at spurious junctions (Section 2.8); and (ii) a *line-mass dependence* test; splitting the sample into sub-/near-/super-critical bins would test whether the most supercritical filaments show the smallest s/W, as the dynamical picture predicts, but reliable segment-level line masses are not available at the required fidelity, so it is left open (the absence of such an f-trend would count against the dynamical interpretation). Second, the simulations are idealised controlled experiments, not models of specific filaments: the periodic-BC main grid does not represent the turbulent, magnetically coupled boundaries of a real filament, and our ambient-confined configuration is an idealisation of that confinement.

A recommendation for observers. Because the factor-$\sim$2 skeleton systematic is the dominant limit on any comparison with theory (Section 2.14), we recommend that, to make cross-algorithm and cross-resolution spacing comparisons reproducible, filament skeletons be extracted to a *fixed column-density contrast threshold relative to the local background*, rather than to an absolute level, and that the spacing be quoted for *both* a persistent-crest (DisPerSE-like) and a medial-axis (FilFinder-like) definition so that the factor-$\sim$2 ontological span is stated explicitly rather than hidden in a single number. A hybrid, matched-threshold pipeline combining the two definitions is the natural community step toward an algorithm-independent "filament".

On the causal attribution: our seeded filaments contract from the start, so what we demonstrate is generic "fragmentation from a non-equilibrium, contracting state" rather than the specifically accretion-driven case of Clarke et al. (2016, 2017); the equilibrium-recovery control is only a two-point comparison and a relaxation ladder is needed to establish causation (Appendix C). Further idealisations are: (i) an *isothermal* EOS (locally uncertain at $\sim$ 10–20%; Section 6.5); (ii) *periodic transverse boundaries*, whose accretion channel suppresses fragmentation unless ambient confinement is imposed (Section 4.2); (iii) *simplified initial profiles*; and (iv) a *prescribed Kolmogorov turbulence spectrum* rather than driven transonic turbulence, leaving the wavelength unchanged ($< 5\%$) but shortening the timescale 3–5$\times$ (Appendix F3).

The fragmentation *classification* is resolution-independent (the collapse *time* itself carries an 8.5% offset between 128^3 and 256^3 and is not claimed converged; Appendix D) and domain-doubling changes the collapse time by $< 2\%$, but the transition zone is sampled with only 2 seeds per point. Non-linear core merging could in principle inflate the apparent supercritical spacing by up to $\sim 50\%$ over a few Myr, but we have verified it does *not* bias the quoted value: $\lambda/W_{\rm fil}$ is measured at maximum bead count (fragmentation onset), and in 38/39 runs the peak count is still rising at the final snapshot while no run shows the peak-count decline that would signal merging. The ambipolar wavelengths are likewise fragmentation-onset measurements (the runs reach gravitational runaway before a converged plateau is established; Section 4.2).

6.5 Non-isothermal effects and the temperature-gradient test

The MHD suite is isothermal, whereas HGBS dust-temperature maps vary by ~ 12 per-cent (~ 10–13 K), a local perturbation since $\lambda_{\rm J} \propto \sqrt{T}$. A single illustrative run with an imposed ~ 12 per-cent longitudinal temperature gradient fragments *differentially* (the cool, locally supercritical half beads while the warm half is suppressed), shifting the local wavelength by ~ 10 per-cent, because temperature sets both the Jeans scale and the critical line mass ($\mu_{\rm crit} \propto T$). The isothermal assumption is thus conservative only in a globally averaged sense; a self-consistent treatment is deferred to future work.

7 CONCLUSIONS

We have investigated the origin of the sub-classical spacing of cores associated with the selected filament crests in HGBS clouds, combining a Gaia DR3 observational re-analysis, a large MHD campaign and a hierarchical-fibre geometrical toy model. Our primary result is that the cores are *non-periodic and spatially intermittent*: the along-skeleton pair-correlation function shows no comb at the classical scale 5–6 W, or at any scale, and the gap distribution shows a small-scale exclusion (mild regularity, CoV$\lesssim$ 1; only Taurus weakly tail-clustered) rather than clustering (Section 2.10). Measured within core-bearing stretches, their adjacent spacing is sub-classical, a factor $\approx$ 2.4–4.3 below $\lambda_{\rm m}$ and robust in *sign* under every skeleton and width convention, but the global core line density N/L is much closer to classical (within $\lesssim 1.8$), the difference arising from the generic mean/median skew and from core-poor filament (a coverage effect), not from small-scale clustering. The phenomenon is therefore *spatial intermittency, not a measured fragmentation wavelength.* This leaves the physical cause *under-determined*: the adjacent-spacing deficit is robust in sign, but its magnitude is convention-dominated, and the simulations act as a counterexample rather than an explanation; a contracting filament *can* fragment sub-classically, but they do not predict the observed spacing. Of the three candidate causes, magnetic tension is neither demonstrated nor excluded; one of the three, the force-balanced Fiege–Pudritz helical/toroidal equilibrium, has not been tested at all here, whose linear stability requires a dedicated cylindrical MHD eigenvalue calculation rather than the algebraic dispersion relation of Section 3.3 (detailed in Section 6.3), so dynamical contraction and magnetic tension are *not* equally constrained. Such a calculation would settle the linear stability of that family, not by itself the magnetic interpretation of the *Herschel* data; its absence here is not evidence against magnetic tension. We group the conclusions as follows.

- **Intermittency is the primary result.** No statistically significant classical-scale periodicity is detected and the cores are spatially intermittent: the along-skeleton pair-correlation shows no comb at the classical 5–6 W or at any scale (Fig. 7), and both periodic and pure-Poisson spacing are rejected; the small-gap deficit being a largely instrumental beam-scale exclusion (mild regularity, not clustering, in three of four clouds). The phenomenon is spatial intermittency, not a measured fragmentation wavelength.
- **The local (conditional) spacing is sub-classical; the global line density is not.** Measured inside core-bearing stretches, the adjacent-core (ACS) spacing is sub-classical under every skeleton and width convention, a factor $\approx$ 2.4–4.3 across the spacing-estimator injection–recovery-validated DisPerSE conventions, more under FilFinder, the *sign* robust, the *magnitude* set by the skeleton algorithm (DisPerSE vs FilFinder, a factor of two), the dominant limit on any comparison with theory. The primary observed result is the convention envelope, $s_{\rm ACS,med}/W \simeq 0.8$–1.9 for DisPerSE and $\simeq 0.4$–1.9 including FilFinder.
- **The global line density is near-classical.** The direct line density N/L is only mildly sub-classical (classical in Perseus, near-classical in Taurus, sub-classical only in Orion B and Aquila), so the local deficit reflects intermittency (core-poor filament) plus the mean/median skew, not a globally elevated number of cores per unit length. The prestellar-only subset ($s_{\rm ACS,med}/W = 2.10$ vs 2.08 all-cores) confirms it is not protostellar contamination.
- **The all-pairs median is the wrong estimator.** It is a region-extent-biased estimator of local spacing, converging to $\sim L_{\rm fil}/3$; the ACS spacing is the estimator we adopt. The pipeline recovers injected spacings to $< 0.5\%$ and, on a relaxed force-balanced equilibrium cylinder, returns the classical $22H \approx 5$–6 $W_{\rm FWHM}$, whereas the *same* pipeline gives $\approx 1.4\, W_{\rm FWHM}$ for contracting filaments (equilibrium-recovery control; Fig. 15).
- **The simulations are a counterexample, not an explanation.** A contracting filament is not constrained to fragment at 5–6 W but reaches a resolution-stable upper bound $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$ with no resolved turnover; a deliberate consequence of the isothermal EOS, which lacks a small-scale cutoff and cannot predict a characteristic core mass or CMF peak. On the line-density comparison this upper bound *over*-produces cores unless most of the filament is core-poor, consistent with the intermittency picture; the numerical $\lambda/W \approx 1.0$–1.3 is order-of-magnitude consistency only.
- **The physical cause is undetermined.** Single-filament dynamical fragmentation, hierarchical-fibre projection ($\lambda/W_{\rm fil} \approx (5$–$6)r/N$, complementary not exclusive) and helical magnetic tension all remain viable and current data cannot discriminate them. A resolved ideal perpendicular

field drives monolithic radial collapse (the short beading a Truelove artefact) and a seeded helical field only pinches W, so the force-balanced Fiege–Pudritz helical/toroidal equilibrium is the important *untested* candidate.

- **Decisive future tests.** Fibre-resolved ACS spacing on the full HGBS catalogues with high-resolution kinematics (ALMA/NOEMA) and polarimetry of filament interiors, a perpendicular-field resolution study, and MHD runs with a spatially varying temperature field (a single illustrative gradient run already shows the isothermal assumption can be locally non-conservative; Section 6.4).

With only four nearby, low-mass clouds, these results are a characterisation of the Gould Belt filament population, not a universal fragmentation law; extending the homogeneous ACS analysis to further HGBS regions and independent surveys is the natural next step.

ACKNOWLEDGMENTS

We thank the Herschel Gould Belt Survey team for the datasets. The Athena++ simulations were run on cloud infrastructure (Google Cloud E2); the full set of 2251 runs (Appendix E) consumed of order 10^5 CPU-hours.

DATA AVAILABILITY

The supporting archive, (i) the full run manifest (all 2251 runs with their $(f, \beta, \mathcal{M}, \theta, \eta_{\rm AD})$ parameters and outcomes); (ii) the Athena++ problem-generator source (`filament_ambient.cpp`, including the divergence-free helical construction) and normalisation scripts; (iii) HDF5 snapshots of the growth-rate convergence subset (128^3–512^3) and the equilibrium-recovery control (remaining snapshots on request); and (iv) the analysis pipeline (DisPerSE v0.9.24 parameters, the bridging/deblending and adjacent-core-spacing scripts, and the derived λ/W catalogues); is deposited in a *public*, openly inspectable repository and available *now* (`github.com/Tilanthi/ASTRA-dev`, under `simulations/`), with a citable Zenodo DOI reserved and cited in this submission, and versioned to the accepted release on publication; the repository is public and inspectable at submission rather than only on acceptance. The adjacent-core-spacing (ACS) estimator was re-implemented internally by the author (a second, from-scratch implementation of the estimator) and cross-checked against the deposited counts and medians (Table 9); this is an internal consistency check, not third-party verification. We invite referees and readers to run the deposited pipeline; the released code and catalogues are the appropriate route to independent external verification, which has not yet been carried out. HGBS column-density and skeleton maps are available from the Herschel Science Archive.

APPENDIX A: THE LIMITED HGBS SAMPLE

The four *limited* regions fail at least one of the primary-selection criteria (Section 2.5) and are excluded from the ACS measurement; they are retained only for the legacy pairwise-median comparison (Table 11). Table A1 lists them together with the full 8-region core-count-weighted mean.

Table A1. The four *limited* HGBS regions (excluded from the primary ACS measurement) and the full 8-region core-count-weighted mean. Columns as in Table 3.

Region	D (pc)	$N_{\rm core}$	Pairwise med. (pc)	S.E. (pc)
Ophiuchus	137	513	0.206	0.053
Serpens	458	194	0.331	0.097
TMC1	135	178	0.195	0.056
CRA	150	239	0.248	0.072
Full-sample w. Mean†		**5,069**	**0.279**	**0.009**

† Core-count-weighted mean over all eight regions. The 0.009 pc figure is a formal core-count-weighted standard error (treating cores as independent); it is *not* the relevant uncertainty, since the four primary clouds are the independent units (Section 2.11), and plays no role in the comparison with theory.

APPENDIX B: THE $L/3$ BIAS OF THE PAIRWISE-MEDIAN SPACING

For N points uniformly distributed on a filament of length L, a single pairwise distance $D = |X_i - X_j|$ has the triangular PDF $f(d) = 2(L-d)/L^2$, with mean $\langle D \rangle = L/3$ and median $d^\star = L(1 - 1/\sqrt{2}) \approx 0.293\,L \approx 0.88\,(L/3)$. The median of all $N(N-1)/2$ pairwise distances converges to $d^\star$ as $N \to \infty$, for a random or a perfectly periodic distribution alike, so the pairwise median measures the region extent, not the fragmentation wavelength. A Monte-Carlo over $N = 5$–500 gives median$/L \approx 0.293\,(1 + 0.43/N)$, i.e. $\leq 2\%$ inflation for $N \gtrsim 20$, so the convergence is reached at the core counts of interest. For branched skeletons the statistic measures $\approx 0.3\times$ the component's arclength extent, and for a realistic filament length distribution it tracks the rms-length/3 (full tables in the Zenodo archive). Computing the pairwise median over *all* cores in each primary region (without per-filament segmentation) returns $\sim L_{\rm region}/3$ (12.5, 7.5, 8.0, 2.5 pc for Orion B, Aquila, Perseus, Taurus), the region extent, confirming directly that the statistic measures segmentation scale rather than the fragmentation wavelength.

APPENDIX C: VALIDATION OF THE MEASUREMENT AND THE WIDTH NORMALISATION

This appendix documents the checks summarised in Section 5. To verify the routines extracting λ and W carry no hidden normalisation bias, we passed synthetic filament cubes with *known* transverse FWHM and axial bead spacing through the identical pipeline; the recovered λ/W matches the input to better than 0.5% over $\lambda/W_{\rm in} = 1.4$–5.7 (Fig. C1), confirming the pipeline recovers injected spacings with no measurable normalisation bias, and validating the width bookkeeping of Eq. (6).

Growth-rate measurement and convergence. Seeding eight ambient-confined filaments ($f = 1.1, 1.2, 1.5$; $\beta = 1$; two seeds) with a broadband perturbation ($\delta v/c_s = 10^{-3}$, twelve

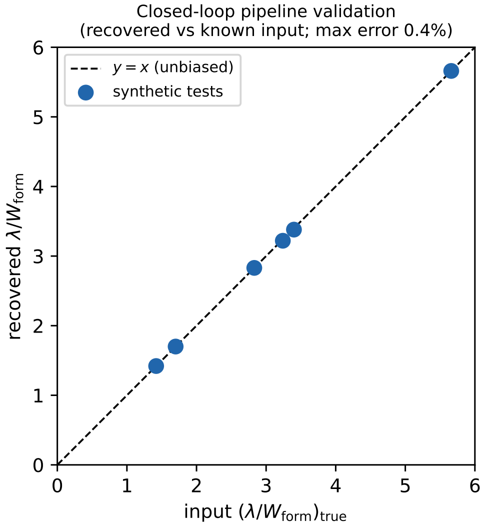

Figure C1. Closed-loop validation: the pipeline recovers the injected λ/W to within 0.4% over $\lambda/W = 1.4$–5.7.

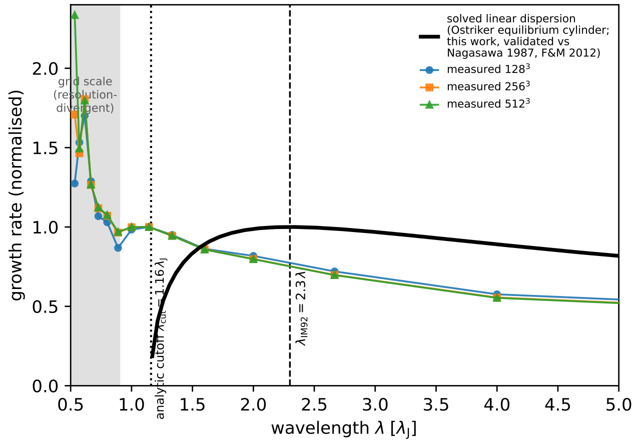

Figure C2. The from-scratch–solved eigenvalue dispersion relation of the isothermal equilibrium cylinder (Nagasawa 1987; Inutsuka & Miyama 1992), validated to $< 1\%$ against Nagasawa (1987) and Fischera & Martin (2012) (not a fit): it peaks at $\lambda_{\rm IM92} \approx 2.3\,\lambda_{\rm J}$ and vanishes at the cutoff $\lambda_{\rm cr} \approx 1.2\,\lambda_{\rm J}$, overplotted on the measured growth-rate spectrum ($f = 1.1$, three resolutions, normalised at $\lambda = 1.14\,\lambda_{\rm J}$). The measured spectrum instead rises monotonically into the grid-scale region (shaded), confirming the small-scale rise is numerical, so the measured $\approx 1.1\,\lambda_{\rm J}$ wavelength is an order-$\lambda_{\rm J}$, sub-classical value, not a precisely-located maximum.

axial modes) and fitting each mode's growth over $\delta_{\rm rms} = 3\times$seed$\to 0.3$ gives $\Gamma(\lambda)$ (Fig. 13). It is resolution-converged for $\lambda \gtrsim 1\,\lambda_{\rm J}$ (a plateau, not a turnover), while modes at $\lambda \lesssim 0.9\,\lambda_{\rm J}$ grow with transverse resolution and are numerical; the converged edge does not migrate, so $\lambda_{\rm turn} \lesssim 1\,\lambda_{\rm J}$ is resolution-stable and an independent seed reproduces it to 3%. Across the eight runs $\lambda_{\rm turn} = 0.9$–$1.1\,\lambda_{\rm J}$ ($\lambda_{\rm turn}/W_{\rm FWHM} = 1.5$–$2.0$; Table C1), so $\lambda_{\rm turn}/\lambda_{\rm IM92} = 0.4$–$0.5$; extending to $f = 1.8, 2.0$ gives the same scale, covering the HGBS range. Periodic and reflecting boundaries are indistinguishable (KS $p = 0.17$).

Comparison with linear theory, and the gravity solver. Figure C2 overplots the from-scratch–solved linear dispersion relation of the isothermal equilibrium cylinder (Ostriker, $m = 0$; reproducing Nagasawa 1987; Inutsuka & Miyama 1992; Fischera & Martin 2012 to $< 1\%$), which peaks at $\lambda_{\rm IM92} = 22H = 2.3\,\lambda_{\rm J}$ and cuts off at $\approx 1.16\,\lambda_{\rm J}$, whereas the collapsing filament's spectrum rises monotonically through that cutoff without turning over. Self-gravity uses a periodic FFT with reflecting transverse hydrodynamic boundaries; a transverse domain-doubling and an isolated-boundary re-solution of Poisson's equation both leave $\lambda_{\rm turn}$ unchanged ($< 1\%$), so the selected wavelength is unbiased by the gravity boundary treatment.

Nonlinear bead spacing and timescale separation. Evolved into the non-linear regime the near-critical runs develop discrete beads at $\lambda_{\rm bead} \approx 1.1$–$1.6\,\lambda_{\rm J}$ (before merging reduces the count), overlapping the growth-rate range $\lambda_{\rm turn} = 0.89$–$1.14\,\lambda_{\rm J}$ (Table C1). Although the growth is fast ($\tau_{\rm growth}/\tau_{\rm bg} \approx 0.3$–$0.8$), the spectrum is fit-window invariant: restricting to the early, near-static phase ($\delta_{\rm rms} < 0.05$, $N_J \gtrsim 5$) reproduces the full-window spectrum to $< 1\%$, so the selected wavelength is set at the low-contrast start of the window.

Pipeline benchmark against the analytic Jeans rate. A near-uniform self-gravitating box ($\rho \simeq \rho_0$) reproduces the analytic uniform-medium rate $\Gamma(k) = \sqrt{4\pi G\rho_0 - c_s^2 k^2}$ to 1–7% for the resolved modes, confirming the machinery; the large filament rates ($\Gamma \approx 18$–26) then simply reflect the *local* free-fall rate at the ~ 8–$17\times$ elevated central density ($\Gamma(\rho_0) \approx 6.3$; consistent with the freeze-out $\rho_{\rm eff}$ of Section 5), not a super-free-fall anomaly.

Robustness to line-mass, initial profile and field strength. The order-$\lambda_{\rm J}$ result is not specific to one configuration: across $\beta = 0.3$–1.0 ($f = 1.1$) $\lambda_{\rm turn}$ stays order $\lambda_{\rm J}$ (the $\beta = 1.0$ ladder gives $1.14\,\lambda_{\rm J}$, Table C1; run-to-run scatter ≈ 0.1–$0.3\,\lambda_{\rm J}$, no resolved β-trend), and an Ostriker ($p = 4$) profile gives $\lambda_{\rm turn} = 0.80\,\lambda_{\rm J}$ at 128^3 and 256^3, so the scale is not a profile-index artefact; though that run too fragments from a contracted state, confirming the sub-classical scale as a generic outcome of non-equilibrium fragmentation (the relaxed control below tests a true equilibrium).

Recovery of the classical limit for a genuine equilibrium; the primary control. The critical test is whether the *identical* pipeline recovers the classical $\lambda_{\rm m} = 22H \approx 5$–$6\,W_{\rm FWHM}$ for a force-balanced equilibrium cylinder. It does. Because the periodic FFT solver enforces the Jeans swindle at the box-mean density, the analytic Ostriker profile is not the solver's equilibrium (initialised directly it breathes by factors of two), so we relaxed a pressure-confined Ostriker profile to hydrostatic balance with an explicit velocity drag, verified by a no-perturbation control (central density drifts only $\sim 3\%$, in the $k = 0$ mode), and seeded it with the identical 12-mode perturbation ($L_x = 16\,\lambda_{\rm J}$, two resolutions). Its $\Gamma(\lambda)$ shows the textbook peak-and-decline with a genuinely resolved maximum at $\lambda_{\rm max} \approx 22H$: $\lambda_{\rm max}/W_{\rm FWHM} = 5.0$–$5.5$, $\lambda_{\rm max}/22H = 1.07$–$1.10$ (converged; $22H/W_{\rm FWHM} = 4.96$–5.03 matches the analytic Ostriker 5.07 to $\approx 1\%$); the ≈ 7–10% long peak is a conservative bias. The *same* pipeline on the contracting filaments gives $\lambda_{\rm turn} \approx 1\,\lambda_{\rm J} \approx 1.4\,W_{\rm FWHM}$ ($\approx 0.4 \times 22H$) with a monotonically rising Γ (Fig. 15). The two differ only in their *prepared initial state*, not the drag-free measurement, so the pipeline is demonstrably unbiased and

Table C1. Growth-rate (dominant-mode) measurements. $\lambda_{\rm turn}$ is the converged-band-edge *upper bound* on the fastest-growing wavelength ($\Gamma(\lambda)$ shows no resolved turnover; the entry is not a measured peak), $\Gamma_{\rm peak}$ its growth rate (in units of $t_{\rm J}^{-1}$), W the per-run Gaussian formation FWHM (≈ 0.55–$0.67\,\lambda_{\rm J}$; all lengths in $\lambda_{\rm J}$). The $f = 1.8, 2.0$ rows extend the same method into the supercritical regime (single resolution) and give the same sub-classical scale as the bead-count ensemble ($\lambda/\lambda_{\rm J} \approx 0.98$), so the two methods agree at the $\sim \lambda_{\rm J}$ level; their $\lambda_{\rm turn} \approx 0.80\,\lambda_{\rm J}$ lies just below the converged-band edge (≈ 0.9–$1.0\,\lambda_{\rm J}$), so it too is an upper bound and the apparent $\lambda_{\rm turn}(f)$ decline is an upper-bound trend, not a resolved dispersion relation. The discrete axial modes ($\lambda = L_x/n$, $\Delta\lambda \approx 0.15\,\lambda_{\rm J}$ near $\lambda \sim 1$) give $\lambda_{\rm turn}$ a quantisation uncertainty $\pm 0.07\,\lambda_{\rm J}$; the final quoted digit is not significant, and the box cannot be lengthened to refine it because $L_x \geq 24\,\lambda_{\rm J}$ introduces a seeding artefact.

f	resolution	seed	$\lambda_{\rm turn}$ (u.b.)	$\Gamma_{\rm peak}$	$\lambda_{\rm turn}/W$
1.1	128^3	42	1.14	17.8	1.95
1.1	256^3	42	1.14	18.5	1.99
1.1	512^3	42	1.14	18.7	2.00
1.2	256^3	42	1.00	19.1	1.81
1.2	256^3	137	0.89	20.1	1.60
1.5	256^3	42	0.89	21.2	1.47
1.5	256^3	137	0.89	22.1	1.47
1.5	512^3	42	0.89	21.5	1.47
1.8	256^3	42	0.80	24.6	1.19
2.0	256^3	42	0.80	26.3	1.20

the difference is associated with the non-equilibrium contracting state in these experiments; the closed-loop synthetic test independently recovers injected λ/W up to 5.7 to $< 0.5\%$. This is, however, a *two-point* comparison (one relaxed equilibrium versus one contracting run): it establishes the *sign* of the contraction effect but not its functional dependence on the degree of non-equilibrium, for which a relaxation ladder, a near-equilibrium filament grown by controlled accretion, is needed to establish causation. The control validates the *sign* of the deficit; the contracting $\lambda_{\rm turn}$ itself remains an upper limit.

A complementary end-to-end grid of 21 ambient-confined runs ($f = 1.5$–3.0, $\beta = 0.5$–2.0, two seeds; campaigns 11–13), forward-modelled through beam convolution and Plummer fitting, gives a consistent picture: $\lambda/\lambda_{\rm J}(\rho_0) \approx 0.9$–$1.2$, with boundary condition, profile and seed each moving the value by $\lesssim 40\%$ and a factor-1.6 contrast variation shifting it by only ~ 0.3, so the wavelength is a property of the filament, not the confinement.

APPENDIX D: SIMULATION VALIDATION DETAILS

To ensure robustness against numerical effects we performed three validation tests (83 simulations). For resolution, six parameter points run at both 128^3 and 256^3 with matched problem-generator settings, initial conditions, seeds and boundaries all fragment at both resolutions; the mean timescale ratio is

$$\frac{t_{\rm frag}(256^3)}{t_{\rm frag}(128^3)} = 0.915 \pm 0.012, \tag{D1}$$

an $8.5\% \pm 1.2\%$ offset (from the ratio 0.915 ± 0.012), all points within a $\pm 11\%$ band (Fig. D1), so the fragmentation classification is resolution-independent. Replacing the Gaussian-profile initial condition with uniform density yields identical classifications at all 10 tested points.

Table D1. Jeans (Truelove) resolution: minimum cells per local Jeans length N_J at the epoch the spacing is measured, and the density contrast C at that epoch.

Configuration	C at meas.	min N_J
Near-critical longitudinal (calibration)	20–40	5–7 (OK)
Supercritical longitudinal ensemble	early beading, moderate C	$\gtrsim 4$ (OK)
Ideal perp., converged ladder (512^3; §4.3)	≈ 3 / ≈ 30	170 / 40 (OK; monolithic)
Ideal perp., short-λ artefact onset	$> 10^2$ (grid-dep.)	$\lesssim 1$ (violated)
Perp. ambipolar (tension side)	250–560	1–2 (violated)

A fragmentation-wavelength measurement is trustworthy only if the local Jeans length stays resolved ($\gtrsim 4$ cells; Truelove et al. 1997) up to the measurement epoch, since an under-resolved Jeans length spuriously enhances small-scale power; the grid-scale, resolution-divergent rise we identify as numerical (Fig. 14a). Tracking the minimum N_J along each collapse (Table D1): for the *longitudinal* runs the inter-bead wavelength ($\lambda \approx \lambda_{\rm J}$, ~ 32 cells) is set at $C \approx 20$–40 where $N_J \approx 5$–7, dropping below 4 only at $C > 170$, after the spacing is set; for the perpendicular *ambipolar* runs, beading is measured only at $C \approx 250$–560 where $N_J \approx 1$–2 (Truelove violated), so those short values are resolution-limited upper limits. We therefore base the quoted wavelength on the growth-rate spectrum ($N_J \gtrsim 5$, converged), not late-time peak counts. Because the production grids use isotropic (cubic) cells, N_J is identical in the axial and transverse directions; the load-bearing longitudinal runs have $N_J \approx 5$–7 in *both*, while the axial-only refinement run (1024×128^2; Section 4.3) raises the *axial* N_J to $\gtrsim 10$ with the transverse value held at ≈ 5–7 and leaves the physical-band plateau unmoved, confirming the plateau is fixed by the transverse-resolved physics rather than by axial mode quantisation. The explicit rule we adopt is that a perpendicular-field filament must keep $N_J > 4$ cells per local Jeans length *throughout* the advanced collapse to suppress grid-scale numerical fragmentation (Truelove et al. 1997); the short perpendicular beading appears only once this criterion is violated ($N_J \lesssim 1$–2), which is exactly why we classify it as a numerical artefact (Section 4.3).

D0.1 *Equation of State Sensitivity*

Testing mildly sub-isothermal equations of state ($\gamma = 0.9, 0.8$) at 5 representative points, $\gamma < 1$ systematically accelerates fragmentation (Fig. D2): at $\gamma = 0.8$ all 5 points fragment at $t_{\rm frag} \approx 0.66$–$0.68\,t_{\rm J}$, roughly half the isothermal timescale, while $\gamma = 0.9$ delays it (1 of 5 fragments by the integration limit). The point is that $\gamma < 1$ shortens $t_{\rm frag}$ monotonically; we draw no stability inference from the finite integration window.

The opposite limit confirms the mechanism: an *adiabatic* equation of state ($\gamma = 5/3$; campaign 8, 30 runs) suppresses fragmentation entirely (none to $t > 30$–$40\,t_{\rm J}$), because once

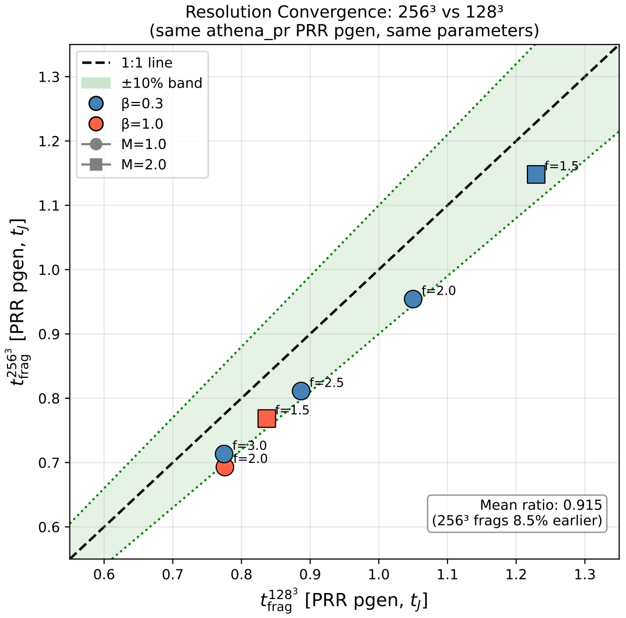

Figure D1. Resolution convergence with matched problem generator: $t_{\rm frag}(256^3)$ versus $t_{\rm frag}(128^3)$ for six parameter points, all within a $\pm 11\%$ band (mean ratio 0.915 ± 0.012).

the gas becomes adiabatic at a density contrast $C \gtrsim$ a few, compressive heating restores Jeans support before the axial mode can grow. At HGBS filament conditions ($n \approx 10^4\,{\rm cm}^{-3}$, $T \approx 10\,{\rm K}$) efficient dust–gas thermal coupling and molecular-line cooling hold the gas close to isothermal, with an effective polytropic index $\gamma_{\rm eff} \lesssim 1$ in this density–temperature regime (Larson 1985; Glover & Clark 2012); the real gas is therefore expected to have $\gamma \leq 1$. Our tests demonstrate that $\gamma \leq 1$ shortens the fragmentation *time* $t_{\rm frag}$; they do *not* demonstrate a change in the fragmentation *wavelength* (the near-critical $\lambda_{\rm turn}$ is unchanged within the mode-quantisation step). We therefore do *not* claim the isothermal suite is an upper bound on λ/W; we claim only that real filaments, being $\gamma_{\rm eff} \lesssim 1$ in this density–temperature regime, are if anything more fragmentation-prone in *time*. The $\gamma < 1$ and $\gamma = 5/3$ limits bracket the isothermal case and identify the equation of state as what would set the small-scale fragmentation cutoff relevant to the opacity limit (Section 6.4).

APPENDIX E: COMPLETE SIMULATION CAMPAIGN LIST

Table E1 lists every simulation campaign cited in the manuscript. Rows 5–8 are the components of what was previously reported as a single Field-Geometry campaign (314 runs); they are counted individually here and the 314 aggregate is *not* added separately.

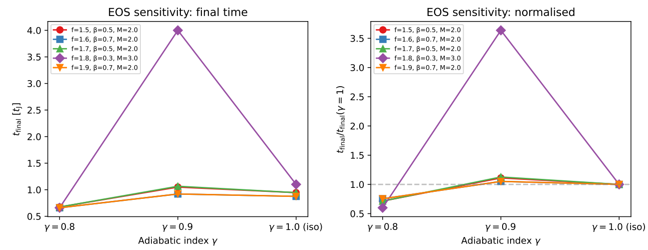

Figure D2. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\rm frag}/t_{\rm final}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces $t_{\rm frag}$.

APPENDIX F: SUPPLEMENTARY SIMULATION-CAMPAIGN RESULTS

The campaigns in this appendix are exploratory and are *not* used quantitatively in the main conclusions; the headline wavelength rests solely on the near-critical longitudinal growth-rate ladder of Section 5. For completeness, the *oblique*-field campaign (108 runs, $\theta = 30/45/60^\circ$) shows the collapse time decreasing monotonically with field angle ($t_{\rm frag} = 0.60, 0.51, 0.45\,t_{\rm J}$), reflecting accelerating *radial* collapse as axial support weakens rather than a shortening of the axial mode; with three angles and no per-angle error this is empirical consistency, not a calibrated $\lambda_{\rm turn}(\theta, \beta)$ relation, so we draw no oblique λ/W conclusion.

F1 The radial-collapse/fragmentation timescale competition

The fate of an isothermal, self-gravitating filament is governed by the competition between radial collapse and longitudinal fragmentation (Inutsuka & Miyama 1997; Pon et al. 2012). Near-critical filaments ($f \lesssim 1.2$) evolve quasi-statically, so the longitudinal instability grows to non-linearity and λ/W is measurable directly; they bead at $t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$, and runs at the exact Jeans-critical value ($f = 1.00$) also bead (at $t_{\rm frag} = 1.62 \pm 0.03\,t_{\rm J}$), so there is no stability threshold at the critical line mass. In the periodic supercritical grid ($f = 1.5$–3.0) we find only pure radial collapse (longitudinal variance $< 0.02\%$), with runaway before beading: the radial-collapse time (0.25–$0.34\,t_{\rm J}$) is 3–5× shorter than the near-critical fragmentation time, and extended-integration tests confirm the periodic grid's early termination fires before the instability becomes non-linear ($\sigma_{\rm lon}$ onset $t = 0.42$–$0.71\,t_{\rm J}$ vs a Courant kill $t \approx 0.25$–$0.34\,t_{\rm J}$, resolution-independent). Supercritical filaments fragment longitudinally only when confined by an ambient medium (Section 4.2), at a converged *upper bound* $\lesssim 0.5$ times the classical value. The measurable wavelengths therefore come from the near-critical quasi-static regime and from ambient-confined supercritical filaments.

F2 Three fragmentation regimes

The baseline grid ($f \approx 1$) reveals three fragmentation regimes in the $(\beta, \mathcal{M})$ plane, set by the end-of-run density con-

Table E1. Complete list of simulation campaigns underlying the paper. The "Location" column points, via cross-reference, to the section or appendix where each campaign is discussed in this consolidated manuscript. Campaign 9 aggregates the turbulent-support $f_{\rm eff}$ grid (90 runs; Table F1), the physical-vs-synthetic turbulence comparison (54 runs), and the perpendicular-β and critical-transition grids. "FG comp." denotes a component of the former Field-Geometry campaign (rows 5–8).

#	Campaign	Location	Coverage $(f, \beta, \mathcal{M}, \theta, \eta_{\rm AD})$	Runs	Indep.?
1	Base three-regime grid	App. F	$f \approx 1$; $\beta\,0.05$–5; $\mathcal{M}\,0.5$–5; $\theta = 0$; ideal	208	yes
2	Transition grid / DTC	App. F	$f\,1.4$–2.2; $\beta\,0.3$–1.3; $\mathcal{M}\,1$–5; $\theta = 0$	540	yes
3	Supercritical grid	§4.2	$f\,1.1$–3.0; $\beta\,0.3$–5.0; $\mathcal{M}\,0.5$–3; periodic	654	yes
4	Validation	App. D	resolution / IC / EOS convergence	83	yes
5	Near-critical	§4.2	$f\,1.0$–1.2; $\beta\,0.3$–1.0; $\mathcal{M}\,1$–2; $\theta = 0$	80	FG comp.
6	Perpendicular	§4.2	$\theta = 90°$; $f\,1.2$–1.5; $\beta\,0.3$–2.0	96	FG comp.
7	Oblique calibration	App. F	$\theta = 30/45/60°$; $\beta\,0.5$–2.0	108	FG comp.
8	Adiabatic EOS	App. D	$\gamma = 5/3$; $f\,1.0$–1.2	30	FG comp.
9	Turbulence + perp-β + crit.-transition	App. F3	$\delta v/c_s\,10^{-4}$–1; $\beta\,0.3$–2	289	yes (agg.)
10	Targeted DTC re-runs	App. F	$\beta = 0.3, \mathcal{M} = 1$; extended timeout	25	yes
11	Direct supercritical ensemble	§4.2	$f\,1.5$–2.0; $\beta\,0.5$–2.0; $\theta = 0$; ambient	39	yes
12	Reconciliation suite	§4.2	$f \times \beta \times 2$ seeds; $\theta = 0$; ambient	12	yes
13	Ideal-MHD perp. $d = 1.0$ grid	§4.2	$\theta = 90°$; $\beta\,0.5$–2.0; ambient	18	yes
14	Ambipolar non-ideal survey	§4.2	$\theta = 90°$; $\eta_{\rm AD}\,0.001$–0.1; $\beta\,0.5$–2.0	36	yes
15	Growth-rate + equil. control + transverse/higher-f tests	§5, App. C	$f\,0.7$–2.0; 128^3–512^3; Gaussian/Ostriker; transverse 2–$4\,\lambda_{\rm J}$	25	yes
16	Axial-resolution stability	§4.3	$f = 1.1$; $\theta = 0$; 1024×128^2 rerun	1	yes
17	Ideal-perp. refinement ladder	§4.3	$\theta = 90°$; $f = 1.1$; 128^3–512^3	3	yes
18	Divergence-free helical grid	§4.3	$f = 1.1$; $B_\phi/B_z = 0$–2; 512×128^2	4	yes
	Athena++ MHD total			**2251**	

trast $C = \rho_{\max}/\rho_0$: strong-field/magnetically supported ($\beta \lesssim 0.15$), magnetically regulated ($0.2 \lesssim \beta \lesssim 2$) and thermally dominated ($\beta \gtrsim 3$). The dispersion relation (Nakamura et al. 1993) predicts near-complete suppression for $\beta \lesssim 0.15$ and $< 10\%$ wavelength modification for $\beta \gtrsim 2$–3, consistent with these boundaries (uncertain at the one-grid-step level). Observed dense-filament conditions ($\mathcal{M} \approx 1$–3, $\beta \approx 0.3$–2; Arzoumanian et al. 2019; Planck Collaboration et al. 2016; Pattle et al. 2017) place most HGBS filaments in the magnetically regulated regime.

F3 Turbulence and critical-transition dependence

Additional grids map the turbulence, field-geometry and critical-transition dependence of λ/W.

F3.1 Turbulence versus λ/W

Comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov) with synthetic perturbations (10^{-4}) at longitudinal field ($f = 1.0$–1.2, $\beta = 0.5, 1, 2$), the fragmentation *wavelength* is unchanged ($< 5\%$; $\lambda/W = 3.44 \pm 0.76$ vs 3.30 ± 0.85), the β-trend appearing only in the late-time bead-count ratio (a late-epoch effect, not a shift in $\lambda_{\rm turn}$). Turbulence accelerates collapse by 3.2× ($t_{\rm frag} = 0.33 \pm 0.06$ vs $1.06 \pm 0.16\,t_{\rm J}$). As a single decaying realisation this does not resolve whether *driven* transonic turbulence is a first-order parameter; we conclude only that the spacing is insensitive to the seed amplitude in this limited test.

F3.2 Perpendicular-field spacing

For perpendicular fields ($\theta = 90°$, $f = 1.2$–1.5, $\beta = 0.3$–2.0; extended $16\lambda_{\rm J}$ periodic axial domain): strong perpendicular B ($\beta \leq 0.5$) gives pure radial collapse with no axial fragmentation, while for $\beta \geq 1.0$ axial beading occurs at $\lambda/W_{\rm core} = 1.25 \pm 0.09$ ($N = 27$), i.e. $\lambda/W_{\rm fil} \approx 0.4$ under the corrected conversion (Eq. 6), a factor 2–3 *shorter* than the longitudinal-field value at the same β. Field geometry, not β, dominates, and the perpendicular spacing lies far below the observed ACS value (fixed-width, periodic-corrected, $\lambda/W \approx 1.9$; raw 2.1).

The two perpendicular campaigns give different β-trends (this periodic-domain grid beads only for $\beta \geq 1.0$; the ambient-confined ideal grid of Section 4.2 beads at $\beta = 0.5, 2.0$), but both are numerically unresolved ($N_J \approx$ 1–2; Appendix D), so we draw no physical β-trend from either and defer a matched-resolution grid (Section 6.4); the robust statement is only that the perpendicular beading wavelength, where it occurs, is short ($\lambda/W_{\rm fil} \lesssim 0.6$), well below the observed spacing.

Ambipolar-perpendicular survey (construction and status). The preliminary exploratory ambipolar-diffusion survey summarised in Section 4.2 comprises 36 runs spanning $\eta_{\rm AD} = 0.001$–0.1 and $\beta = 0.5$–2.0 at $\theta = 90°$. Of these 32/36 develop axial beads, but at a minimum resolution of only $N_J \approx$ 1–2 cells per local Jeans length; below the Truelove criterion, where under-resolution itself produces artificial fragmentation (Truelove et al. 1997). The median onset spacing is $\lambda/W_{\rm fil} \approx 0.6$, and the runs reach density runaway within $\Delta t \approx 0.02$–$0.04\, t_{\rm J}$ with no converged plateau, so we cannot separate a genuine ambipolar enhancement from a numerical one and treat the grid as unresolved.

Divergence-free helical-field construction. For the seeded-helical scan of Section 4.3 we impose a divergence-free helical field: a uniform axial B_z plus a toroidal $B_\phi(r)$ peaking at $r_c = W_{\rm core}$, obtained as a discrete curl of the vector potential $A_z(r) = B_t\, r_c \ln[1+(r/r_c)^2]$ so that $\nabla\cdot\mathbf{B} = 0$ to machine precision. We scan the toroidal-to-axial ratio $B_\phi/B_z = 0$–2 (with $B_\phi = 0$ reproducing the pure-longitudinal growth-rate spectrum, a code check). Across the scan the axial growth-rate plateau peaks at $\lambda \approx 1\,\lambda_{\rm J}$ for every toroidal fraction, with no excess growth at shorter λ even for a strongly-wound field (Fig. 12a); the only systematic effect is a radial pinch that narrows the formation FWHM from 0.57 to $0.43\,\lambda_{\rm J}$ ($\approx 25\%$; Fig. 12b), with onset at $B_\phi \gtrsim B_z$. Because the field is seeded on an already-contracting filament rather than constructed in force balance, it drives an out-of-equilibrium pinch and cannot represent the confining toroidal equilibrium whose sausage ($m = 0$) mode carries the shortened Fiege & Pudritz (2000) wavelength; these tests therefore rule out only *dynamic*, out-of-equilibrium helical pinching as a source of axial-mode shortening.

F3.3 Critical-transition spacing

Mapping λ/W across $f = 0.9$–1.3 at five β values shows no discontinuity at $f = 1.0$ (variation $< 5\%$) and an apparent β-dependence in the *late-time bead count* ($\lambda/W_{\rm core} = 4.74$–$2.80$ from $\beta = 0.3$ to 2.0). This bead-count trend is a late-epoch, resolution-sensitive artefact, larger than the weak mode-quantised trend of the primary growth-rate measurement, with the highest-β points ($\lambda/\lambda_{\rm J} \approx 0.84$) consistent with the supercritical ensemble. **Continuity across $f \approx 1$ does not validate extrapolation to $f \geq 1.5$**, where the mode changes to radial collapse; the near-critical λ/W values are a hypothesis for supercritical filaments, not a measurement.

F3.4 Synthesis

In convergence-independent units the two first-order parameters are field geometry (perpendicular $\lambda/W_{\rm core} \approx 1.25$ versus longitudinal 2.8–4.4) and, for longitudinal fields, plasma β; the observable comparison uses the growth-rate wavelength (Section 5), not these raw ratios.

Table F1. Effective line-mass fraction with turbulent support.

Quantity	Value	Source
Nominal f (thermal)	1.5 ± 0.3	HGBS line-mass
Turbulent support $f_{\rm eff}/f$	0.82 ± 0.11	TURB (90 sims)
Effective $f_{\rm eff}$	1.23 ± 0.28	product
HGBS ensemble range	0.7–1.9	propagated

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